Regional Influenza Outbreak Severity Prediction for Healthcare Planning

University of San Diego

ADS 503 Applied Predictive Modeling

Celina Velazquez, Yubi Joy Quinzon , Sheshma Jaganathan

Date: 6/22/2026

https://github.com/celina-velazquez/ADS-503

Abstract

Seasonal influenza places significant strain on healthcare systems by increasing patient volume, laboratory testing demand, and resource utilization during outbreak periods. The objective of this study was to develop a predictive model capable of forecasting influenza outbreak severity in advance to support proactive healthcare planning and preparedness.

This analysis utilized historical influenza surveillance data obtained from the Centers for Disease Control and Prevention (CDC), including the ILINet and Clinical Laboratory datasets spanning 2015 through 2026. Following data cleaning and integration, feature engineering was performed to create additional predictors that captured historical influenza trends, recent activity patterns, and seasonality. To support real-world forecasting, all engineered features were constructed using only information available prior to the prediction period, preventing data leakage and improving the validity of model evaluation.

Exploratory data analysis was conducted to better understand influenza activity patterns and identify variables associated with future outbreak severity. The analysis revealed strong seasonal trends, geographic variation in influenza activity, and temporal persistence in outbreak behavior, suggesting that historical surveillance measures may provide meaningful signals for forecasting future influenza positivity.

Multiple predictive modeling approaches, including Ridge Regression, Random Forest, and XGBoost, were evaluated using cross-validation and a time-ordered train-test split to reflect real-world forecasting conditions. Several engineered features, including previous-week influenza positivity, rolling averages, and laboratory surveillance measures, demonstrated strong relationships with future influenza activity.

The results demonstrate that historical influenza surveillance data can provide meaningful early warning signals of future outbreak activity. These forecasts may help healthcare organizations improve staffing decisions, resource allocation, laboratory preparedness, and overall response planning during periods of elevated influenza activity.

Contents

Introduction

Problem Statement

2.1 Problem Statement and Background

Seasonal influenza remains a significant healthcare challenge in the United States, contributing to increased healthcare demand and resource utilization during annual outbreak periods (CDC, 2024). During peak flu seasons, hospitals and clinics often experience increased patient visits, higher testing demand, and greater use of healthcare resources. Because influenza activity can vary across regions and time periods, it can be difficult for healthcare organizations to prepare for sudden increases in cases.

The Centers for Disease Control and Prevention (CDC) monitors influenza activity through surveillance systems such as ILINet and Clinical Laboratory reporting networks. These systems track influenza-like illness (ILI) reported by healthcare providers and laboratory-confirmed influenza cases. While this information is useful for understanding current influenza conditions, it mainly describes what is happening now and provides limited insight into future outbreak severity.

To address this challenge, this study uses historical CDC surveillance data to develop a regression-based predictive model for forecasting influenza positivity one week ahead across different regions. By identifying patterns in past influenza activity, the model provides early warning signals of increasing influenza transmission and potential outbreak severity. These forecasts can help healthcare organizations improve preparedness, plan resources more effectively, and respond proactively to periods of elevated influenza activity.

2.2 Business Questions

To support healthcare preparedness and resource planning, the analysis focuses on the following business questions:

- Which regions are expected to experience the highest influenza activity and may require early healthcare preparedness?
- Can we predict which weeks are likely to experience peak influenza intensity that may increase pressure on healthcare systems?
- Which historical patterns and surveillance indicators provide early warning signals of increasing influenza activity?
- Which surveillance variables are the strongest predictors of 1 week ahead influenza positivity ?

Methodology

3.1 Data Understanding

This project uses two influenza surveillance datasets obtained from the Centers for Disease Control and Prevention (CDC) FluView surveillance portal. The first dataset, ILINet (Influenza-Like Illness Network), contains 30,154 weekly observations of influenza-like illness activity reported by healthcare providers across U.S. states and territories between 2015 and 2026. The second dataset, Clinical Laboratory Surveillance, contains 29,810 weekly laboratory testing records, including total specimens tested, confirmed Influenza A cases, confirmed Influenza B cases, and influenza test positivity rates.

Combining these datasets provides a comprehensive view of influenza activity by integrating symptom-based surveillance data with laboratory-confirmed influenza testing information across regions and time periods.

Key predictor variables include: *region, year, week, percent_unweighted_ili, ilitotal, num_of_providers, total_patients, total_specimens, total_a, total_b*.

The target variable is target_1wk, which represents the 1-week-ahead influenza positivity rate. It was feature engineered by shifting *percent_positive* forward by one week within each region. The variable serves as a measure of future influenza outbreak severity and is used as the response variable for regression modeling.

3.2 Data Preparation and Preprocessing

Column names were standardized using a consistent naming convention to improve readability and reduce coding errors. Variables containing suppressed CDC values ("X") were converted to missing values and evaluated for completeness. Variables with excessive missingness, including *percent_weighted_ili* and all age-group variables, were removed because they contained no usable information.

Missing values were handled using a combination of:

- Linear interpolation for percentage-based surveillance variables
- Forward-fill and backward-fill techniques for time-series continuity
- Median imputation for count variables such as *ilitotal*, *total_patients*, *total_specimens*, *total_a*, and *total_b*

Data were sorted by *region, year,* and *week* to preserve temporal ordering before imputation. Outlier analysis using the IQR method identified extreme observations in variables such as *total_a, total_b, total_specimens,* and *percent_positive*. These values were retained because they likely represent genuine influenza outbreaks and periods of elevated healthcare demand rather than data quality issues.

After preprocessing and merging the two CDC datasets using *region, year*, and *week*, the final analytical dataset contained 27,601 observations and 11 variables, producing a unified influenza surveillance dataset suitable for predictive modeling.

3.3 Feature Engineering

Feature engineering was performed to create additional predictors that capture temporal, seasonal, and epidemiological patterns in influenza activity. Because influenza outbreaks evolve over time rather than occurring independently each week, historical surveillance information was used to construct features that provide context about recent disease trends and outbreak progression.

An additional objective of the feature engineering process was to prevent data leakage and ensure that all predictors reflected information that would realistically be available at the time a forecast is generated. Because the goal of this study is to predict future influenza activity, engineered features were constructed using only historical observations. Lagged variables and rolling averages were calculated exclusively from prior weeks' data, while future influenza positivity values were reserved as target variables rather than predictors. This approach helps ensure that model performance more accurately reflects real-world forecasting conditions and avoids artificially optimistic results.

To support forecasting objectives, four future target variables were created by shifting influenza positivity rates forward by one to four weeks within each region. These variables represent future influenza activity and allow the modeling process to evaluate short-term outbreak prediction performance at multiple forecasting horizons.

Several lagged variables were created to capture influenza conditions from the previous week, including influenza positivity, influenza-like illness (ILI) activity, laboratory testing volume, and confirmed Influenza A and B cases. These variables were included because current influenza activity is often influenced by patterns observed in prior weeks.

Rolling averages based on the previous four weeks were also calculated for key surveillance variables. Rolling averages help smooth short-term fluctuations and noise while preserving broader outbreak trends. These features provide a measure of recent influenza activity that may be more stable than single-week observations.

To capture week-to-week changes in influenza conditions, difference variables were created for laboratory testing counts, influenza positivity, and ILI activity. These variables measure the rate of increase or decrease in influenza activity and may serve as early indicators of emerging outbreaks.

Several ratio variables were engineered to represent relationships between surveillance measures. These included the proportion of laboratory specimens identified as Influenza A, the proportion identified as Influenza B, and the ratio of ILI cases to total patient visits. These features provide normalized measures of influenza activity that are less sensitive to differences in population size or testing volume across regions.

Finally, seasonal features were created using sine and cosine transformations of the calendar week variable. Influenza activity follows a recurring seasonal cycle, and these cyclical transformations allow the model to recognize that adjacent weeks at the beginning and end of the year are more closely related than a simple numeric week value would suggest.

Together, these engineered features were designed to capture historical influenza trends, seasonal effects, outbreak momentum, and relative disease activity. By incorporating both raw surveillance measures and derived temporal indicators, the feature engineering process aimed to improve the model's ability to predict future influenza positivity and outbreak severity.

3.4 Feature Engineering Validation and Cleanup

Following feature engineering, the newly created variables were evaluated to ensure they were suitable for predictive modeling and did not introduce unintended data quality issues. Missing value analysis was performed on all engineered features to identify observations that could not be calculated due to insufficient historical data, unavailable future observations, or undefined mathematical operations.

A portion of the missing values resulted naturally from the feature engineering process. Lagged variables and rolling averages require historical observations, causing the first several weeks within each region to contain incomplete values. Similarly, the future target variables required for forecasting could not be calculated for the final weeks of each regional time series because future influenza observations were unavailable. These missing values were expected and reflected the temporal structure of the data.

Additional investigation revealed that the engineered ratio variables contained missing values caused by division-by-zero scenarios. Specifically, observations with zero laboratory specimens produced undefined values for the Influenza A and Influenza B ratio variables, while observations with zero patient visits produced undefined values for the ILI patient ratio. Because these missing values were generated by valid surveillance conditions rather than missing data, the affected ratio variables were imputed with a value of zero to preserve the associated observations.

After addressing ratio-based missing values, the remaining incomplete observations were removed to ensure a fully populated modeling dataset. This step produced a complete analytical dataset containing only observations with valid predictor and target values. The resulting dataset provided a consistent foundation for exploratory analysis, feature selection, model training, and performance evaluation

3.5 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand influenza activity patterns, evaluate data quality, and identify factors related to influenza outbreak severity.

Summary statistics, as shown in Table 1, showed that variables such as *percent_positive, percent_unweighted_ili, ilitotal, total_specimens, total_a,* and *total_b* varied considerably across regions and weeks. Most count variables were right-skewed, meaning that very high influenza activity occurred during only a small number of weeks, while most observations represented normal influenza conditions. The distribution of *percent_positive* was also right-skewed, indicating that high influenza positivity rates occurred during relatively few outbreak periods.Because the target variable, *target_1wk*, is derived from *percent_positive* ,a similar pattern is expected in influenza positivity levels.This suggests that periods of elevated influenza activity are relatively uncommon and tend to occur during specific outbreak seasons.

Table 1

Summary Statistics of Key Influenza Variables

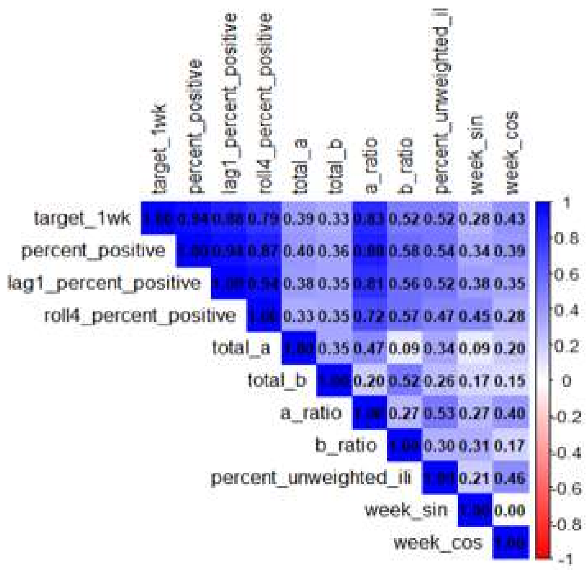
	n	mean	sd	min	max	skew
percent_positive	27600	5.52	8.65	0	66.67	2.03
percent_unweighted_ili	27600	2.14	2.02	0	19.33	2.23
ilitotal	27600	774.17	1625.26	0	26030.00	5.61
total_patients	27600	30258.92	40860.39	0	317409.00	3.40
total_specimens	27600	986.76	1946.21	0	36660.00	6.05
total_a	27600	71.55	322.82	0	9364.00	13.21
total_b	27600	19.34	85.97	0	2054.00	10.32

To reduce skewness and improve the distribution of highly skewed count variables, a log transformation (log1p) was applied to variables such as *ilitotal,percent_positive, total_patients, total_specimens, total_a, total_b,* and their corresponding lagged and rolling-average features. The transformation substantially reduced skewness and the distributions became more symmetric and suitable for regression modeling.

Correlation Analysis - Several variables showed strong positive correlations with target_1wk, the target variable (Figure 1). The strongest relationships were observed for percent_positive (r = 0.94), lag1_percent_positive (r = 0.94), roll4_percent_positive (r = 0.88), and a_ratio (r = 0.83). Laboratory surveillance variables, including total_a (r = 0.39) and total_b (r = 0.33), showed moderate positive correlations, while percent_unweighted_ili (r = 0.52) was also moderately associated with the target. These findings suggest that recent influenza activity, represented by current and lagged positivity rates, is a stronger predictor of future influenza activity than laboratory testing volumes alone.

Figure 1

Top Variables Correlated with Target Variable

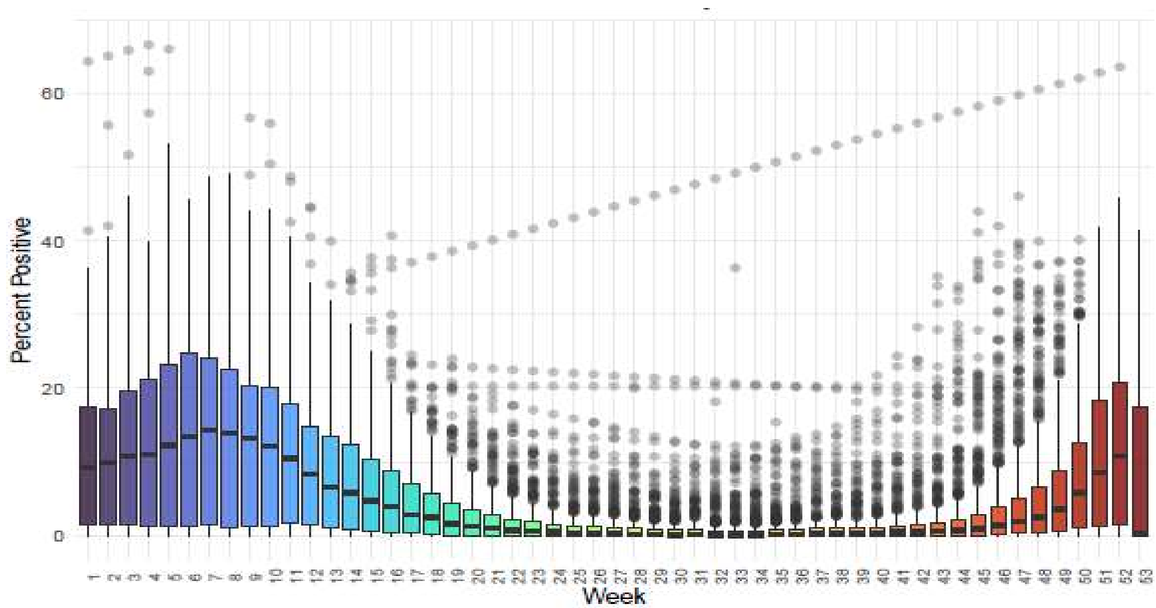


Geographic Analysis - Geographic analysis of historical influenza activity (2015–2026) revealed differences across states. Wyoming (14.14%), Louisiana (7.64%), Nebraska (7.61%), Idaho (7.55%), and Texas (7.53%) had some of the highest average influenza positivity rates. These results indicate that influenza activity varies across regions, with some states consistently experiencing higher influenza burden than others.

Seasonal Analysis - Figure 1 shows how influenza positivity varies across different weeks throughout the year**.** Influenza positivity generally increased during late fall and winter, peaked during the traditional influenza season (approximately weeks 1–10 and weeks 48–52), and remained low during most summer weeks.

Figure 2

Distribution of Influenza Positivity Across Weeks of the Year



This pattern confirms the strong seasonal nature of influenza activity, generally increasing during the fall and winter months and decreasing during the spring and summer months.

In addition to seasonal timing, influenza strain activity was also examined to determine which strain contributed most to outbreak peaks. The comparison of Influenza A *(total_a)* and Influenza B (*total_b*) during annual peak outbreak weeks showed that Influenza A cases were consistently higher than Influenza B cases. This suggests that Influenza A contributed more to seasonal outbreak peaks over the study period. Understanding which strain is more prevalent may help support influenza surveillance and preparedness efforts.

Persistence Analysis - The persistence analysis showed that low-activity weeks were most frequently followed by another low-activity week (20,693 observations). Similarly, High activity weeks were often followed by another High activity week (1,981 observations). This indicates that influenza activity tends to persist from one week to the next, with activity levels changing gradually over time rather than abruptly. These findings support the use of lagged features for outbreak prediction.

The EDA identified important seasonal, geographic, and temporal patterns in influenza activity and highlighted several variables that may be useful for predicting future outbreak severity.

3.6 Modeling

3.6.1 Feature Selection

Influenza data is known to exhibit multicollinearity. To account for this while keeping the highly predictive variables that were created during feature engineering, the original predictors from which features were engineered were removed given their high correlation with the engineered features.

3.6.2 Data Splitting and Resampling Strategies

To create the training and test sets, instead of a stratified random sample or a simple random sample, a time-ordered sample was taken so that future observations did not leak into the training set given the time-series nature of influenza data and impact the reported test performance. Next, the predictor set was created by removing all four response variables. The target variable, target_1wk, was identified. Lastly, cross-validation folds were created where k=5 to reduce computation time. Multiple folds were tested and five-folds were chosen due to computation constraints during model training.

3.6.3 Model Training and Tuning

For linear model evaluation, Ridge Regression was used due to its ability to handle highly correlated predictors which are often present in influenza data, and for its quick training ability. Random Forest was also chosen for its ability to reduce overfitting and for its high predictive accuracy. Lastly, XGBoost was selected for its ability to prevent overfitting and capture nonlinear relationships in the data.

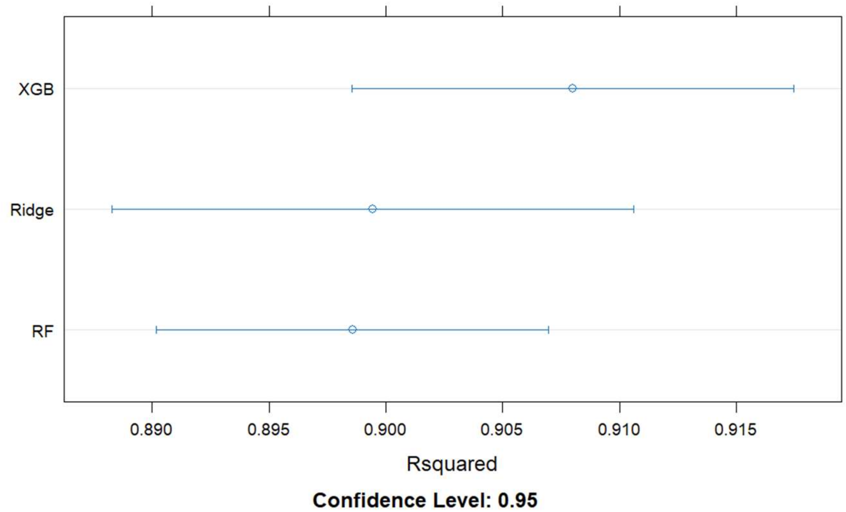
For the ridge regression model, any remaining predictors that were character data types were converted to numeric using one-hot encoding because the model requires numeric predictors. The tuning parameter lambda, which denotes regularization strength, tested more than 30 values from 0.0005 to 0.05. For the Random Forest model, multiple values of the ntree hyperparameter, number of trees, were selected to optimize compute time. In addition, mtry, the number of predictors that the random forest randomly considers at each split in the decision tree, was set to the square root of the number of predictors. Lastly, for the XGBoost model, the number of boosting iterations (nrounds), the max depth of each tree (max_depth), and the learning rate (eta) each had two values evaluated during hyperparameter tuning. These values were 100 and 200 for nrounds, 3 and 6 for max_depth, and 0.05 and 0.1 for eta.

3.6.4 Validation and Testing

A check for statistically significant differences in the RMSE values of each of the models reveals that there is a significant difference in RMSE between XGB and Ridge (p-value = 0.00004) and between XGB and RF (p-value = 0.020). The difference in the prediction error between Ridge and RF (p-value = 0.232) was not statistically significant. XGB achieved the lowest RMSE (2.384) and highest R-squared with a value of 0.925, and, as a result, was selected as the preferred model for prediction.

Figure 2

Comparison of Model Performance Using R^2 (95% CI)



Results

Among the three models evaluated, XGBoost achieved the strongest predictive performance, producing the lowest prediction error and the highest explanatory power (R^2 = 0.922). Statistical testing confirmed that XGBoost significantly outperformed both Ridge Regression ($p = 0.00037$) and Random Forest ($p = 0.01060$), while no significant performance difference was observed between Ridge Regression and Random Forest.

In addition to achieving the strongest predictive performance, XGBoost identified several important predictors for future influenza activity. Variables related to recent influenza positivity rates, Influenza A laboratory counts, rolling historical averages, and influenza-like illness activity were among the most influential features! The importance of these variables suggests that influenza outbreaks exhibit strong temporal persistence, where recent disease activity provides valuable information about near-term outbreak severity. These findings also indicate that laboratory surveillance measures and historical influenza trends can serve as meaningful early warning signals for forecasting future influenza positivity and supporting proactive healthcare planning

Constraints and Future Work

Future work may include adding additional predictors such as regional weather variables or population density metrics which may help to improve performance of the model. In addition, while the model performed well with the predictors used, additional research can be done to potentially eliminate predictors and create a more parsimonious model if the One-Standard-Error Rule is followed where a simpler model with fewer predictors may be chosen if its performance is within one standard error of the best model.

Conclusion

This study developed and evaluated predictive models to forecast regional influenza outbreak severity using CDC surveillance data from 2015 through 2026. Through data preprocessing, feature engineering, exploratory data analysis, and model evaluation, the results demonstrated that historical influenza surveillance data can be used to accurately predict future influenza positivity rates. XGBoost emerged as the preferred model, achieving superior predictive performance and explaining approximately 92% of the variation in future influenza activity.

The findings suggest that healthcare organizations can use predictive analytics to gain advance insight into potential influenza surges, allowing for more proactive staffing, resource allocation, laboratory preparedness, and operational planning. By transforming routinely collected surveillance data into actionable forecasts, this approach has the potential to improve healthcare readiness and support more informed decision-making during periods of elevated influenza activity

References

Kuhn, M., & Johnson, K. (2013). Applied predictive modeling. New York: Springer.

Centers for Disease Control and Prevention. (2024). *FluView interactive: Influenza surveillance data*. U.S. Department of Health and Human Services. https://gis.cdc.gov/grasp/fluview/fluportaldashboard.html

Centers for Disease Control and Prevention. (2024). *Overview of influenza surveillance in the United States*. U.S. Department of Health and Human Services. https://www.cdc.gov/fluview/overview/index.html

OpenAI. (2026). *ChatGPT* (June 2026 version) [Large language model]. https://chatgpt.com

Regional Influenza Outbreak Severity Prediction for Healthcare Planning

The ILINet and Clinical Labs datasets were downloaded from the CDC FluView surveillance system, which provides weekly influenza monitoring data across the United States. The ILINet dataset contains information about influenza-like illness (ILI) activity reported by healthcare providers, including the percentage of flu-like patient visits, total ILI cases, number of providers, and total patient visits.

The Clinical Labs dataset contains laboratory-confirmed influenza testing results, including total specimens tested, confirmed Influenza A and B cases, and positivity percentages. Combining these two datasets helps create a more complete influenza surveillance system by connecting symptom-based flu activity with confirmed laboratory influenza cases for outbreak analysis and prediction.

KEY VARIABLES

Table 1: Key Variables Used in the Analysis

Variable	Description
region	U.S. state or reporting region
year	Surveillance year
week	CDC surveillance week
percent_unweighted_ili	Percentage of patient visits classified as influenza-like illness (ILI)
ilitotal	Total number of influenza-like illness cases reported
num_of_providers	Number of healthcare providers reporting data
total_patients	Total number of patient visits reported
total_specimens	Total number of laboratory specimens tested for influenza
total_a	Number of laboratory-confirmed Influenza A cases
total_b	Number of laboratory-confirmed Influenza B cases
percent_positive	Percentage of specimens testing positive for influenza (Target Variable)

```r
# Load required libraries
library(tidyverse)  #For data manipulation
library(GGally) #visualization
library(janitor) #data manipulation
library(lubridate) #For date handling
library(corrplot) #visualization
library(ggplot2) #visualization
library(scales) ##formatting
library(knitr) #formatting
library(gridExtra) #display plots in grid layout
library(patchwork) #combine multiple ggplot
library(zoo) #imputation and interpolation
library(slider) # Used to help calculate rolling averages
library(e1071) #to calculate skewness
library(dplyr) #data filtering, grouping,summarization
library(usmap) #create U.S. state maps
```

```r
library(sf)#Spatial data processing
library(psych) #exploring data
library(tidyr) # Data reshaping
library(here) #for file paths
library(randomForest) #for random forest modeling
library(caret) #Model evaluation and training
library(xgboost) #for xgboost modeling
library(viridis) #Color palettes
#install.packages("xgboost")
```

IMPORT DATASETS

```r
# CDC datasets contain an extra descriptive row,
# therefore we use skip = 1  during import.
#Reading all columns as character strings to prevent formatting issues
# Import ILINet dataset
ilinet <- read_csv(
  here("FluViewPhase2Data", "ILINet.csv"),
  skip = 1,
  col_types = cols(.default = col_character())
)

clinical_labs <- read_csv(
  here("FluViewPhase2Data", "ICL_NREVSS_Clinical_Labs.csv"),
  skip = 1,
  col_types = cols(.default = col_character())
)
```

VERIFY IMPORTS

```r
#view few rows and columns of ilinet dataset
head(ilinet[,1:7]) |> knitr::kable()
```

REGION TYPE	REGION	YEAR	WEEK	% WEIGHTED ILI	%UNWEIGHTED ILI	AGE 0-4
States	Alabama	2015	40	X	2.58875	X
States	Alaska	2015	40	X	0.627353	X
States	Arizona	2015	40	X	1.23609	X
States	Arkansas	2015	40	X	1.14799	X
States	California	2015	40	X	1.38064	X
States	Colorado	2015	40	X	0.51656	X

```r
#view few rows for clinical lab dataset
head(clinical_labs) |>knitr::kable()
```

REGION TYPE	REGION	YEAR	WEEK	TOTAL SPECIMENS	TOTAL A	TOTAL B	PERCENT POSITIVE	PERCENT A	PERCENT B
States	Alabama	2015	40	167	2	3	2.99	1.2	1.8
States	Alaska	2015	40	X	X	X	X	X	X
States	Arizona	2015	40	55	0	0	0	0	0
States	Arkansas	2015	40	26	0	1	3.85	0	3.85
States	California	2015	40	683	2	0	0.29	0.29	0
States	Colorado	2015	40	255	0	1	0.39	0	0.39

```r
#view dimensions of both datasets
cat("Dimensions for ilinet dataset is ", dim(ilinet))
```

```
## Dimensions for ilinet dataset is  30152 15
```

```r
cat("\n Dimensions for clinical labs dataset is",dim(clinical_labs))
```

```
##
##  Dimensions for clinical labs dataset is 29808 10
```

Output show 30152 rows, 15 columns for ilinet and 30152 rows and 10 columns and 29808 rows for clinical _labs.

VIEW DATATYPES

```r
# inspect structure of both datasets
str(ilinet)
```

```
## spc_tbl_ [30,152 x 15] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ REGION TYPE      : chr [1:30152] "States" "States" "States" "States" ...
##  $ REGION           : chr [1:30152] "Alabama" "Alaska" "Arizona" "Arkansas" ...
##  $ YEAR             : chr [1:30152] "2015" "2015" "2015" "2015" ...
##  $ WEEK             : chr [1:30152] "40" "40" "40" "40" ...
##  $ % WEIGHTED ILI   : chr [1:30152] "X" "X" "X" "X" ...
##  $ %UNWEIGHTED ILI  : chr [1:30152] "2.58875" "0.627353" "1.23609" "1.14799" ...
##  $ AGE 0-4          : chr [1:30152] "X" "X" "X" "X" ...
##  $ AGE 25-49        : chr [1:30152] "X" "X" "X" "X" ...
##  $ AGE 25-64        : chr [1:30152] "X" "X" "X" "X" ...
##  $ AGE 5-24         : chr [1:30152] "X" "X" "X" "X" ...
##  $ AGE 50-64        : chr [1:30152] "X" "X" "X" "X" ...
##  $ AGE 65           : chr [1:30152] "X" "X" "X" "X" ...
##  $ ILITOTAL         : chr [1:30152] "272" "10" "290" "46" ...
##  $ NUM. OF PROVIDERS: chr [1:30152] "28" "7" "52" "14" ...
##  $ TOTAL PATIENTS   : chr [1:30152] "10507" "1594" "23461" "4007" ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   .default = col_character(),
##   ..   `REGION TYPE` = col_character(),
##   ..   REGION = col_character(),
```

```
##   ..   YEAR = col_character(),
##   ..   WEEK = col_character(),
##   ..   `% WEIGHTED ILI` = col_character(),
##   ..   `%UNWEIGHTED ILI` = col_character(),
##   ..   `AGE 0-4` = col_character(),
##   ..   `AGE 25-49` = col_character(),
##   ..   `AGE 25-64` = col_character(),
##   ..   `AGE 5-24` = col_character(),
##   ..   `AGE 50-64` = col_character(),
##   ..   `AGE 65` = col_character(),
##   ..   ILITOTAL = col_character(),
##   ..   `NUM. OF PROVIDERS` = col_character(),
##   ..   `TOTAL PATIENTS` = col_character()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

```r
str(clinical_labs)
```

```
## spc_tbl_ [29,808 x 10] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ REGION TYPE     : chr [1:29808] "States" "States" "States" "States" ...
##  $ REGION          : chr [1:29808] "Alabama" "Alaska" "Arizona" "Arkansas" ...
##  $ YEAR            : chr [1:29808] "2015" "2015" "2015" "2015" ...
##  $ WEEK            : chr [1:29808] "40" "40" "40" "40" ...
##  $ TOTAL SPECIMENS : chr [1:29808] "167" "X" "55" "26" ...
##  $ TOTAL A         : chr [1:29808] "2" "X" "0" "0" ...
##  $ TOTAL B         : chr [1:29808] "3" "X" "0" "1" ...
##  $ PERCENT POSITIVE: chr [1:29808] "2.99" "X" "0" "3.85" ...
##  $ PERCENT A       : chr [1:29808] "1.2" "X" "0" "0" ...
##  $ PERCENT B       : chr [1:29808] "1.8" "X" "0" "3.85" ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   .default = col_character(),
##   ..   `REGION TYPE` = col_character(),
##   ..   REGION = col_character(),
##   ..   YEAR = col_character(),
##   ..   WEEK = col_character(),
##   ..   `TOTAL SPECIMENS` = col_character(),
##   ..   `TOTAL A` = col_character(),
##   ..   `TOTAL B` = col_character(),
##   ..   `PERCENT POSITIVE` = col_character(),
##   ..   `PERCENT A` = col_character(),
##   ..   `PERCENT B` = col_character()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

From above output we can see all variables are currently stored as character strings which will require numeric conversion for analysis. In CDC influenza datasets, “X” usually means the value was suppressed, missing, or not reported.

CLEAN COLUMN NAMES

Original column names contain capital letters,spaces,special characters,and periods.So we Standardize names to improves code readability,

```r
# View column names for ilinet
colnames(ilinet) |>knitr::kable()
```

x
REGION TYPE
REGION
YEAR
WEEK
% WEIGHTED ILI
%UNWEIGHTED ILI
AGE 0-4
AGE 25-49
AGE 25-64
AGE 5-24
AGE 50-64
AGE 65
ILITOTAL
NUM. OF PROVIDERS
TOTAL PATIENTS

```r
#View column names fr Clinical lab dataset
colnames(clinical_labs) |>knitr::kable()
```

x
REGION TYPE
REGION
YEAR
WEEK
TOTAL SPECIMENS
TOTAL A
TOTAL B
PERCENT POSITIVE
PERCENT A
PERCENT B

```r
#To improve code readability and prevents errors we standardize column names to a consistent format
#Convert column names into standardized format
ilinet <- clean_names(ilinet)
#Verify cleaned column names
colnames(ilinet) |>knitr::kable()
```

x
region_type
region
year
week
percent_weighted_ili
percent_unweighted_ili

x
age_0_4
age_25_49
age_25_64
age_5_24
age_50_64
age_65
ilitotal
num_of_providers
total_patients

```
#similarly for cilinal labs dataset
clinical_labs <- clean_names(clinical_labs)
#Verify cleaned column names
colnames(clinical_labs) |>knitr::kable()
```

x
region_type
region
year
week
total_specimens
total_a
total_b
percent_positive
percent_a
percent_b

VIEW DUPLICATES

```
#calculate duplicate for ilinet
duplicate_ili <- sum(duplicated(ilinet))
duplicate_ili
```

```
## [1] 0
```

```
#calculate duplicate for clinical labs
duplicate_clinical <- sum(duplicated(clinical_labs))
duplicate_clinical
```

```
## [1] 0
```

No duplicate records exist in either datasets.

HANDLE SUPPRESSED VALUES X

```r
#"X" usually means the value was suppressed, missing, or not reported.
#R cannot  analyze "X" as numeric data.
#Converts CDC suppression values "X" to proper NA values
ilinet[ilinet == "X"] <- NA
clinical_labs[clinical_labs == "X"] <- NA

#counts missing data per column
missing_ilinet <- colSums(is.na(ilinet))
missing_ilinet |>knitr::kable()
```

	x
region_type	0
region	0
year	0
week	0
percent_weighted_ili	30141
percent_unweighted_ili	592
age_0_4	30152
age_25_49	30152
age_25_64	30152
age_5_24	30152
age_50_64	30152
age_65	30152
ilitotal	592
num_of_providers	592
total_patients	592

```r
missing_clinical <- colSums(is.na(clinical_labs))
missing_clinical |>knitr::kable()
```

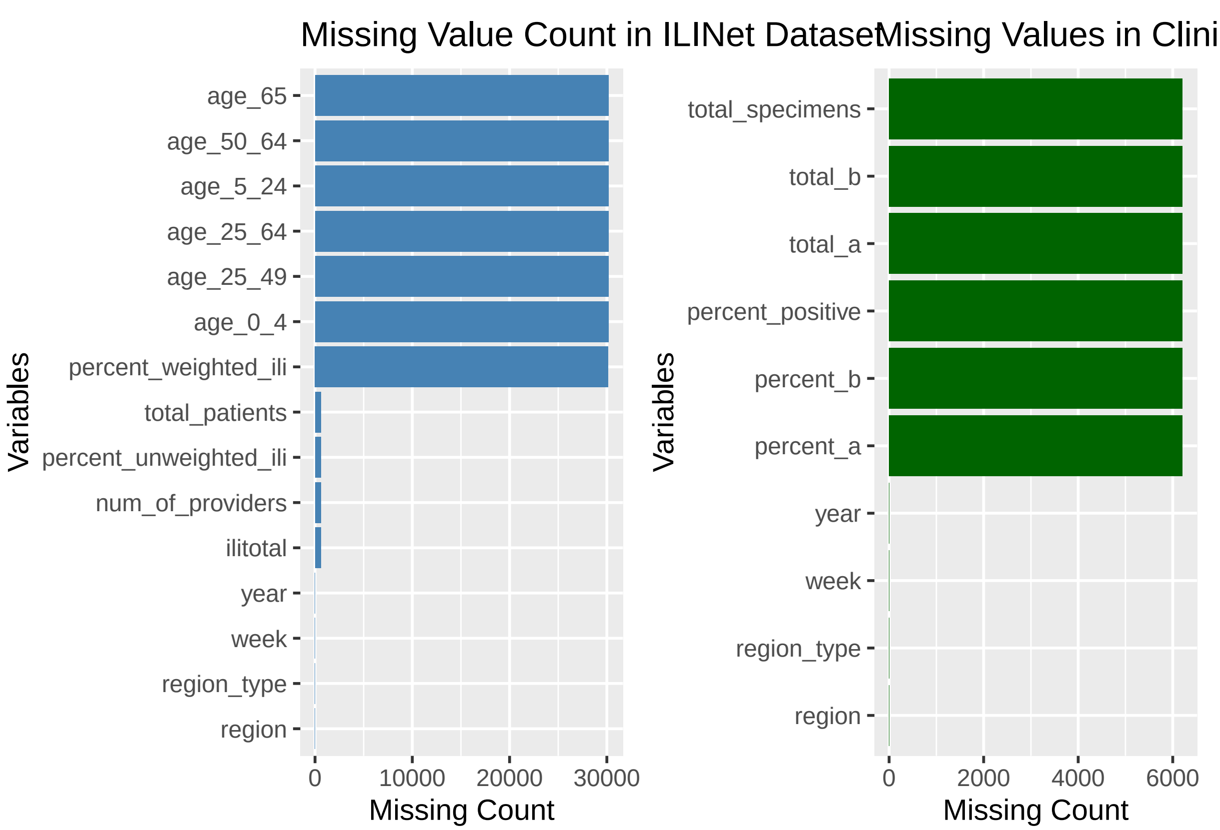
	x
region_type	0
region	0
year	0
week	0
total_specimens	6205
total_a	6205
total_b	6205
percent_positive	6205
percent_a	6205
percent_b	6205

- The missing value analysis shows that percent_weighted_ili contains 30,141 missing values, meaning almost all records in this variable are unavailable and the column may not be reliable for analysis.
- All age-specific variables (age_0_4, age_25_49, age_25_64, age_5_24, age_50_64, and age_65) contain 100% missing values making these variables completely unusable.
- The Clinical Labs dataset contains 6,205 missing values.
- Since the age-group variables contain no valid observations, these columns should be removed during preprocessing to reduce noise and improve overall data quality for influenza outbreak analysis.

MISSING VALUES

We can calculate and visualise the missing value using ggplot for both datasets.

```
# This converts the missing value counts into a dataframe
missing_df_ili <- data.frame(variable = names(missing_ilinet),missing_count = missing_ilinet)
## The bar chart helps identify missingness in the ILINet.
p1 <- ggplot(
  missing_df_ili,
  aes(x = reorder(variable, missing_count),y = missing_count)) +
  geom_bar(stat = "identity",fill = "steelblue" ) + coord_flip() +
  labs(title = "Missing Value Count in ILINet Dataset",x = "Variables",y = "Missing Count")

#Similarly convert missing values into dataframe for clinical labs dataset
missing_df_clinical <- data.frame( variable= names(missing_clinical),missing_count =missing_clinical)
#visualise using barplot
p2 <- ggplot(missing_df_clinical,
  aes(x = reorder(variable, missing_count),y = missing_count)) +
  geom_bar(stat = "identity",fill = "darkgreen") +coord_flip() +
  labs(title = "Missing Values in Clinical Labs",x = "Variables",y = "Missing Count")

#visualize both plots
p1 + p2
```



DROP COMPLETELY MISSING COLUMNS

```r
#Columns with entirely missing values are removed
# Age columns and weighted_ili have 100% missing values so we drop them
ilinet <- ilinet %>%
  select(-starts_with("age_"), -percent_weighted_ili , -region_type)
#similarly drop for clinical labs dataset
clinical_labs <- clinical_labs %>% select(-region_type )

# Verify columns removed
colnames(ilinet) |> knitr::kable()
```

x
region
year
week
percent_unweighted_ili
ilitotal
num_of_providers
total_patients

```r
colnames(clinical_labs) |> knitr::kable()
```

x
region
year
week
total_specimens
total_a
total_b
percent_positive
percent_a
percent_b

After dropping columns, the ILINet dataset is reduced from 15 to 7 meaningful variables.

CONVERT VARIABLES TO NUMERIC

Convert date components, ILI percentages, and count variables to proper numeric types.

```r
#convert numeric variables from character datatype into integer/numeric format.
ilinet <- ilinet %>%
  mutate(
    year = as.integer(year),
    week = as.integer(week),
    percent_unweighted_ili = as.numeric(percent_unweighted_ili),
    ilitotal = as.numeric(ilitotal),
    num_of_providers = as.numeric(num_of_providers),
```

```r
    total_patients = as.numeric(total_patients))

#similarly for clinical_labs
clinical_labs <- clinical_labs %>%
  mutate(
    year = as.integer(year),
    week = as.integer(week),
    total_specimens = as.numeric(total_specimens),
    total_a = as.numeric(total_a),
    total_b = as.numeric(total_b),
    percent_positive = as.numeric(percent_positive),
    percent_a = as.numeric(percent_a),
    percent_b = as.numeric(percent_b)
  )
#Verify datatype
sapply(ilinet, class) |>knitr::kable()
```

	x
region	character
year	integer
week	integer
percent_unweighted_ili	numeric
ilitotal	numeric
num_of_providers	numeric
total_patients	numeric

```r
sapply(clinical_labs, class) |>knitr::kable()
```

	x
region	character
year	integer
week	integer
total_specimens	numeric
total_a	numeric
total_b	numeric
percent_positive	numeric
percent_a	numeric
percent_b	numeric

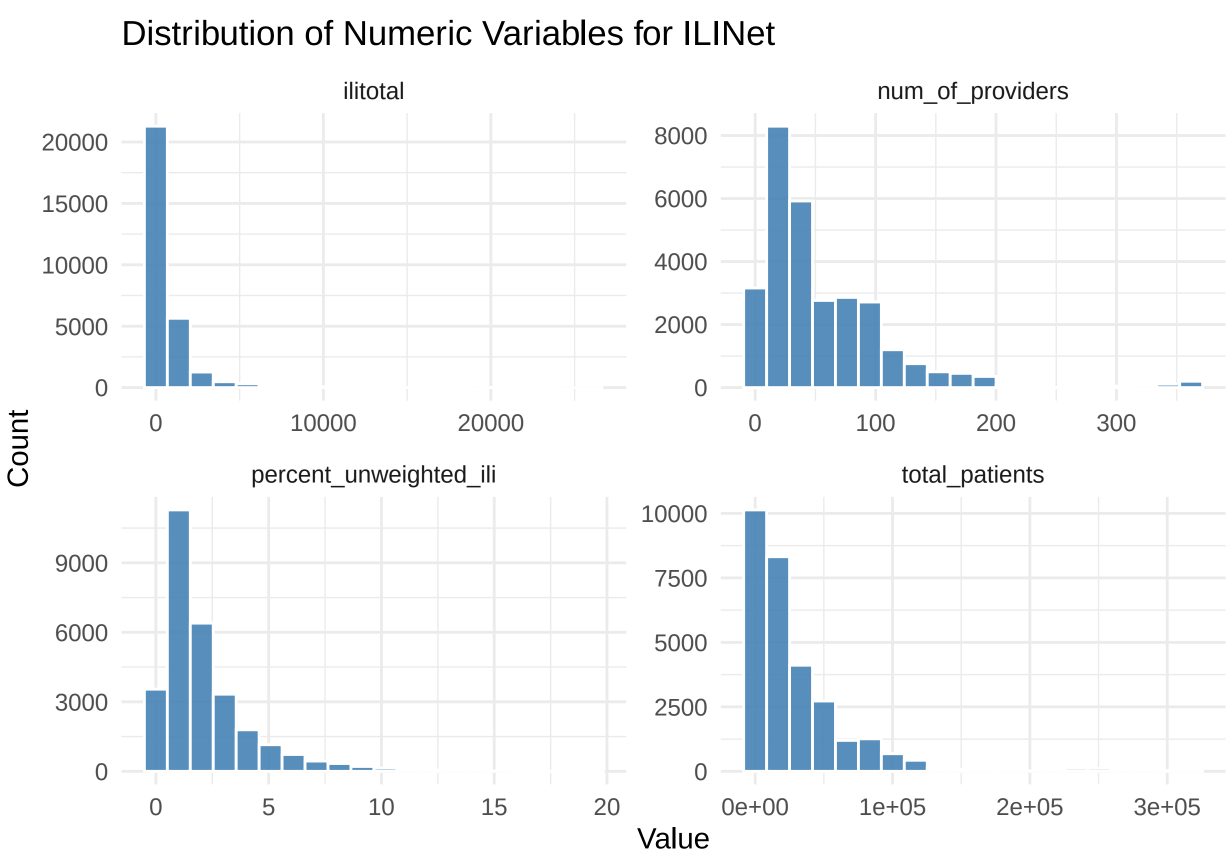
After conversion, approximately 7 variables in ILINet and 8 in clinical labs become numeric.

NUMERIC VARIABLE DISTRIBUTIONS

ILINET DATASET

- Distribution of numeric variable for ilinet dataset

```r
# Create distribution plots for key numeric variables only
#select numeric variables for ilinet dataset
numeric_ilinet <- ilinet %>% select(where(is.numeric),-week,-year)

#Convet all numeric columns into one common structure(long format) so ggplot create multiple panels
numeric_ilinet_long <- numeric_ilinet %>%
  pivot_longer(cols = everything(),names_to = "Variable",values_to = "Value")

#Create histogram distributions for all numeric variables
ggplot(numeric_ilinet_long,aes(x = Value)) +

  geom_histogram(bins = 20,fill ="steelblue",color= "white",alpha = 0.9) +

  facet_wrap(~ Variable,scales = "free",ncol = 2) +   #creates separate plots for each variable

  labs(title = "Distribution of Numeric Variables for ILINet",
    x = "Value",y = "Count") + theme_minimal()
```

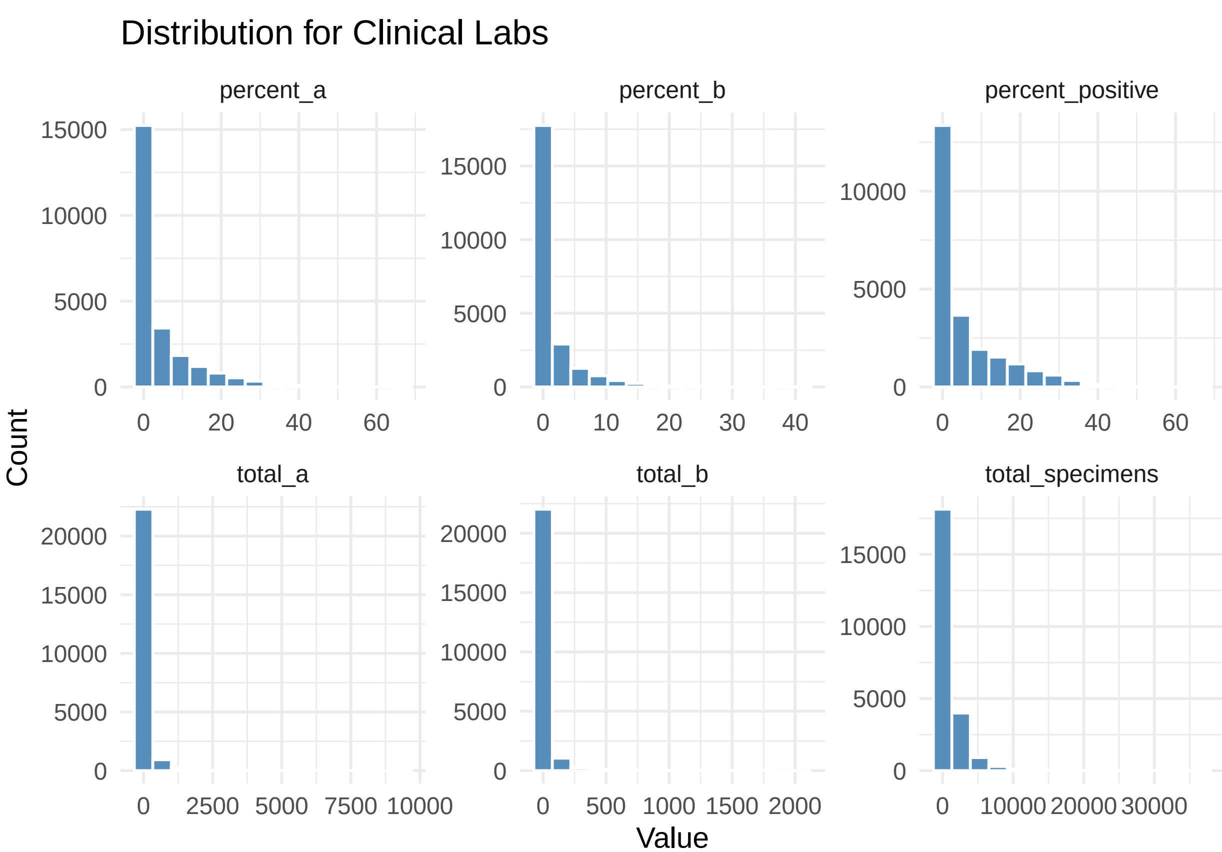


- Most numeric variables are right-skewed.Variables such as ilitotal, total_patients, total_specimens, total_a, and total_b show strong positive skewness.
- percent_unweighted_ili is also positively skewed, indicating that high influenza activity occurs less frequently.
- This pattern is expected because most regions experience normal flu activity, while only a few weeks show major outbreak spikes.
- week and year appear more uniformly distributed.

CLINICAL LABS DATASET

- Distribution of numeric variables for Clinical dataset

```
# Create distribution plot for clinical dataset
#select numeric variables for clinical labs dataset
numeric_clinical <- clinical_labs %>% select(where(is.numeric), -week, -year)

#Convert all numeric columns into one common structure
numeric_clinical_long <- numeric_clinical %>%
  pivot_longer(cols = everything(),names_to = "Variable",values_to = "Value")

#Create histogram distributions for all numeric variables
ggplot(numeric_clinical_long,aes(x = Value)) +
  geom_histogram(bins = 15,fill = "steelblue",color = "white",alpha = 0.9) +

  facet_wrap(~ Variable,scales = "free",ncol = 3) +   #creates separate plots for each variable

  labs(title = "Distribution for Clinical Labs",
    x = "Value",y = "Count") + theme_minimal()
```



- The Clinical Labs numeric variables are highly right-skewed.Variables such as total_specimens, total_a, and total_b show many low laboratory counts with a small number of very large testing volumes.
- percent_positive, percent_a, and percent_b are also positively skewed, indicating that high influenza positivity rates occur less frequently.

OUTLIER DETECTION

```r
# detect outliers using IQR method
ili_vars <- names(numeric_ilinet)
clinical_vars <- names(numeric_clinical)
detect_outliers_iqr <- function(data, var_name) {
  x <- data[[var_name]]
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  IQR <- Q3 - Q1
  lower_bound <- Q1 - 1.5 * IQR
  upper_bound <- Q3 + 1.5 * IQR

  outliers <- which(x < lower_bound | x > upper_bound)

  data.frame(
    Variable = var_name,
    Q1 = Q1,
    Q3 = Q3,
    IQR = IQR,
    Lower_Bound = lower_bound,
    Upper_Bound = upper_bound,
    Outlier_Count = length(outliers),
    Outlier_Percent = length(outliers) / sum(!is.na(x)) * 100
  )
}

# Detect outliers for ILINet
ili_outliers <- bind_rows(lapply(ili_vars, function(v) detect_outliers_iqr(ilinet, v)))

# Detect outliers for Clinical Labs
clinical_outliers <- bind_rows(lapply(clinical_vars, function(v) detect_outliers_iqr(clinical_labs, v)))

# Display results
cat("ILINET OUTLIER DETECTION (IQR Method) \n")
```

```
## ILINET OUTLIER DETECTION (IQR Method)
```

```r
knitr::kable(ili_outliers, digits = 2, row.names = FALSE)
```

Variable	Q1	Q3	IQR	Lower_Bound	Upper_Bound	Outlier_Count	Outlier_Percent
percent_unweighted_ili	0.83	2.76	1.93	-2.06	5.65	2028	6.86
ilitotal	66.00	782.00	716.00	-1008.00	1856.00	2952	9.99
num_of_providers	18.00	78.00	60.00	-72.00	168.00	1194	4.04
total_patients	5271.00	40361.75	35090.75	-47365.12	92997.88	1739	5.88

```r
cat("\n CLINICAL LABS OUTLIER DETECTION (IQR Method) \n")
```

```
##
##  CLINICAL LABS OUTLIER DETECTION (IQR Method)
```

```r
knitr::kable(clinical_outliers, digits = 2, row.names = FALSE)
```

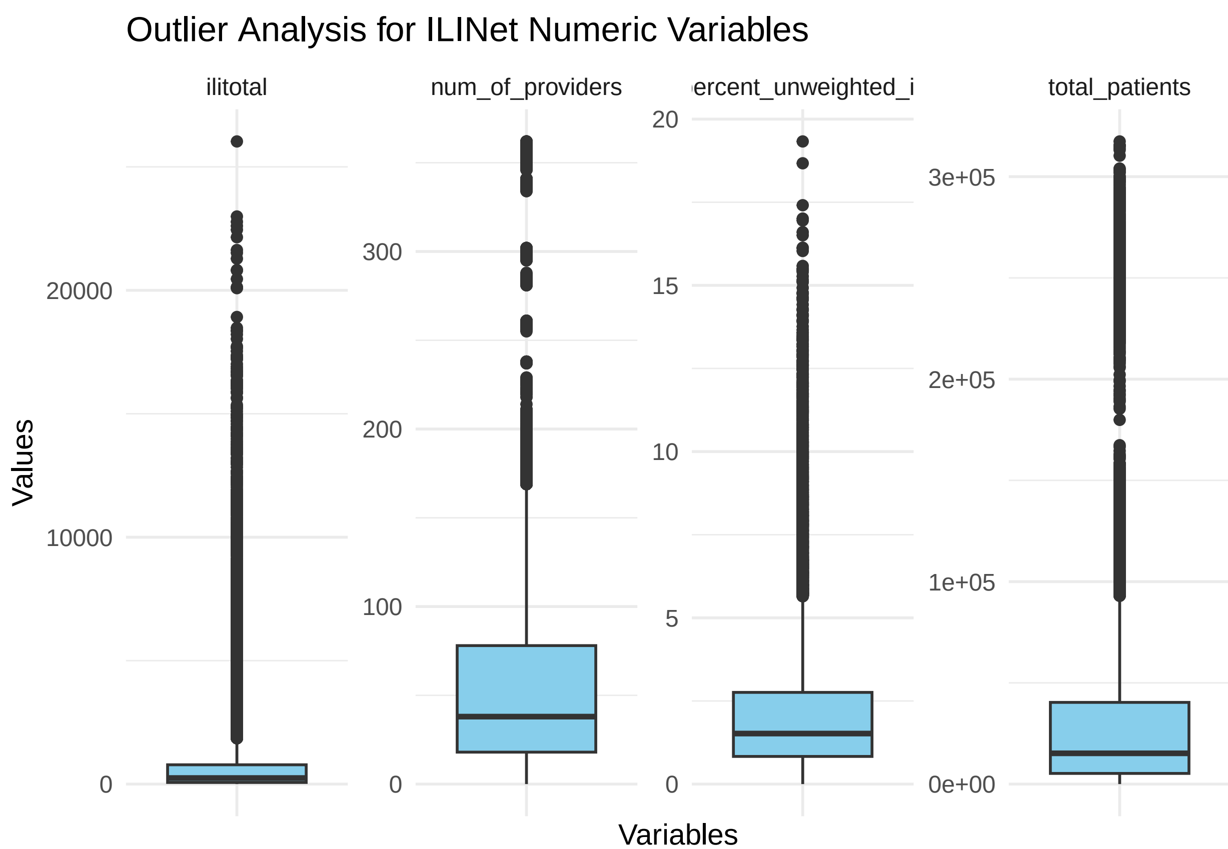
Variable	Q1	Q3	IQR	Lower_Bound	Upper_Bound	Outlier_Count	Outlier_Percent
total_specimens	160.00	1202.00	1042.00	-1403.00	2765.00	2349	9.95
total_a	0.00	34.00	34.00	-51.00	85.00	3779	16.01
total_b	0.00	8.00	8.00	-12.00	20.00	3755	15.91
percent_positive	0.09	8.79	8.70	-12.95	21.83	1895	8.03
percent_a	0.00	5.34	5.34	-8.01	13.35	2660	11.27
percent_b	0.00	1.43	1.43	-2.14	3.58	3481	14.75

- In the ILINet dataset, ilitotal has the highest number of outliers (2,952 records, 9.99%), followed by percent_unweighted_ili (6.86%) and total_patients (5.88%).
- num_of_providers has fewer outliers (4.04%), while week and year show no outliers because they follow fixed calendar ranges.
- In the Clinical Labs dataset, total_a (16.01%), total_b (15.91%), and percent_b (14.75%) contain the highest proportion of outliers.
- total_specimens and percent_positive also show notable outlier percentages, indicating sudden spikes in influenza testing and positivity rates.
- These extreme values represent severe influenza outbreaks, seasonal surges, or regions with higher healthcare demand rather than data entry errors.
- Since outbreak spikes are meaningful public health events, removing all outliers may lead to loss of important influenza activity patterns.
- So instead of deleting them, highly skewed variables can be transformed using log or Box-Cox methods to improve data distribution and model performance.

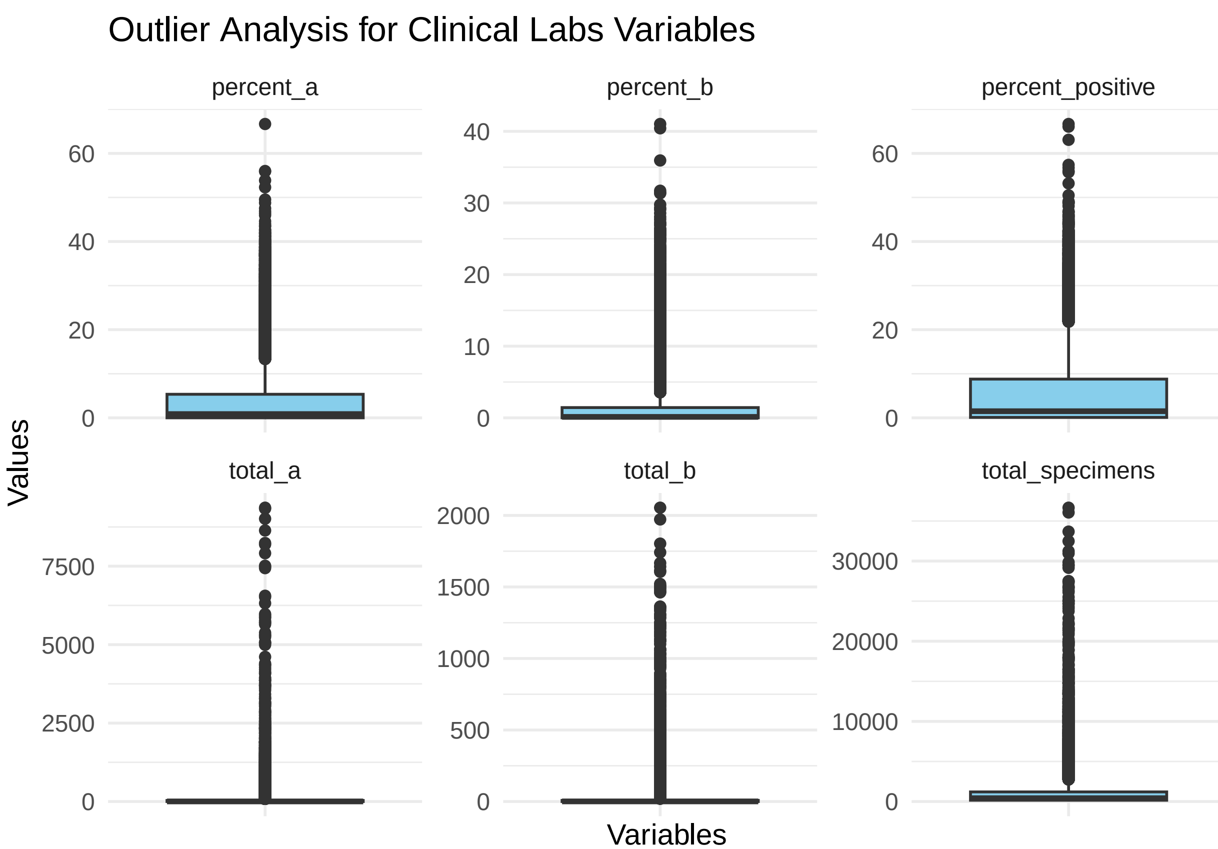
VISUALIZE OUTLIERS DISTRIBUTION

Visualize the outlier distribution for both datasets

```r
# Visualize outliers for ilinet datset
ggplot(numeric_ilinet_long,
  aes(x = Variable,y = Value)) + geom_boxplot(fill = "skyblue") +

  facet_wrap(~ Variable,scales = "free",ncol = 4) +

  theme_minimal() + theme(axis.text.x = element_blank()) +

  labs(title = "Outlier Analysis for ILINet Numeric Variables",x = "Variables",y = "Values")
```




```r
#Similarly visualize outliers for Clinical Labs dataset


ggplot(numeric_clinical_long,
  aes(x = Variable,y = Value)) + geom_boxplot(fill = "skyblue") +

  facet_wrap(~ Variable,scales = "free",ncol = 3) +

  theme_minimal() +theme(axis.text.x = element_blank()) +

  labs(title = "Outlier Analysis for Clinical Labs Variables",x = "Variables",y = "Values")
```



SUMMARY STATISTICS

```r
# View summary statistics for both datasets to understand data quality
summary(numeric_ilinet) |>knitr::kable()
```

percent_unweighted_ili	ilitotal	num_of_providers	total_patients
Min. : 0.0000	Min. : 0.0	Min. : 0.00	Min. : 0
1st Qu.: 0.8315	1st Qu.: 66.0	1st Qu.: 18.00	1st Qu.: 5271
Median : 1.5202	Median : 247.0	Median : 38.00	Median : 15203
Mean : 2.1653	Mean : 785.4	Mean : 55.66	Mean : 30136
3rd Qu.: 2.7587	3rd Qu.: 782.0	3rd Qu.: 78.00	3rd Qu.: 40362
Max. :19.3284	Max. :26030.0	Max. :362.00	Max. :317409
NA's :592	NA's :592	NA's :592	NA's :592

```r
summary(numeric_clinical) |>knitr::kable()
```

total_specimens	total_a	total_b	percent_positive	percent_a	percent_b
Min. : 0	Min. : 0.00	Min. : 0.00	Min. : 0.000	Min. : 0.00	Min. : 0.000
1st Qu.: 160	1st Qu.: 0.00	1st Qu.: 0.00	1st Qu.: 0.090	1st Qu.: 0.00	1st Qu.: 0.000

	total_specimens	total_a	total_b	percent_positive	percent_a	percent_b
	Median : 421	Median : 4.00	Median : 1.00	Median : 1.500	Median : 0.85	Median : 0.130
	Mean : 1114	Mean : 83.11	Mean : 22.52	Mean : 5.976	Mean : 4.33	Mean : 1.645
	3rd Qu.: 1202	3rd Qu.: 34.00	3rd Qu.: 8.00	3rd Qu.: 8.785	3rd Qu.: 5.34	3rd Qu.: 1.430
	Max. :36660	Max. :9364.00	Max. :2054.00	Max. :66.670	Max. :66.67	Max. :41.030
	NA's :6205	NA's :6205	NA's :6205	NA's :6205	NA's :6205	NA's :6205

- In the ILINet dataset, percent_unweighted_ili has a mean of 2.17 and a maximum value of 19.33,showing occasional severe influenza activity spikes.
- ilitotal and total_patients show very large maximum values (26,030 and 317,409),which shows the presence of highly populated regions or major outbreak periods.
- The mean values for ilitotal, num_of_providers, and total_patients are higher than their medians, indicating right-skewed distributions and the influence of outliers.
- In the Clinical Labs dataset, total_specimens, total_a, and total_b also show large maximum values,meaning sudden increases in laboratory testing during outbreak periods.
- percent_positive has a maximum value of 66.67%, showing that some regions experienced extremely high influenza positivity rates during peak seasons.
- The variables week and year are evenly distributed because the data spans multiple influenza seasons and calendar weeks.
- Both datasets contain missing values, particularly in laboratory-related variables,so imputation has to be done before predictive modeling.

IMPUTATION OF MISSING VALUES

- The datasets are sorted by region, year, and week to preserve the natural time-series order of influenza activity before performing imputation.
- Since influenza activity changes gradually over time within each region, interpolation (na.approx) was used to estimate missing percentage-based variables (percent_unweighted_ili and percent_positive) using nearby weekly observations.
- Forward fill (na.locf) and backward fill (fromLast = TRUE) were applied to handle missing values at the beginning or end of weekly sequences where interpolation alone could not estimate values.
- Median imputation was used for ilitotal, total_patients, num_of_providers, total_specimens, total_a, and total_b because the median is less sensitive to extreme outbreak spikes and outliers than the mean.
- Grouping by region was done within each geographic region, preserving regional influenza patterns and preventing data leakage across locations.

```r
# ILINET IMPUTATION

#Sort by region ,year and week  for ilinet dataset
ilinet <- ilinet %>% arrange(region,year,week)

# Impute the missing values
ilinet_clean <- ilinet %>% group_by(region) %>%
  mutate(
    percent_unweighted_ili =
      na.approx(percent_unweighted_ili,na.rm = FALSE,rule = 2),   #forward fill
    percent_unweighted_ili =
      na.locf(percent_unweighted_ili,na.rm = FALSE),
    percent_unweighted_ili =
      na.locf(percent_unweighted_ili,fromLast = TRUE, na.rm = FALSE),
```

```r
    # Median is robust to outbreak spikes
    ilitotal =
      ifelse(is.na(ilitotal),median(ilitotal, na.rm = TRUE),ilitotal),
    total_patients =
      ifelse(
        is.na(total_patients),median(total_patients, na.rm = TRUE),total_patients),
    num_of_providers =
      ifelse(is.na(num_of_providers),median(num_of_providers, na.rm = TRUE),
        num_of_providers)) %>%  ungroup()

# Verify ilinet datset after imputaion
colSums(is.na(ilinet_clean)) |> kable()
```

	x
region	0
year	0
week	0
percent_unweighted_ili	344
ilitotal	344
num_of_providers	344
total_patients	344

```r
# CLINICAL LABS IMPUTATION
#Similar steps are done for clinical labs dataset
# Sort data by region ,year and week

clinical_labs <- clinical_labs %>% arrange(region,year,week)

# Impute missing values for clinical labs

clinical_labs_clean <- clinical_labs %>% group_by(region) %>%
  mutate(
    percent_positive =na.approx(percent_positive,na.rm = FALSE,rule = 2),
    percent_positive =na.locf(percent_positive,na.rm = FALSE),
    percent_positive =na.locf(percent_positive,fromLast = TRUE,na.rm = FALSE),
    total_specimens =
      ifelse(is.na(total_specimens),median(total_specimens, na.rm = TRUE),
        total_specimens),
    total_a =
      ifelse(is.na(total_a),median(total_a, na.rm = TRUE),total_a ),
    total_b =
      ifelse(
        is.na(total_b),median(total_b, na.rm = TRUE),total_b)) %>% ungroup()


# Verify clinical valyes missing values
colSums(is.na(clinical_labs_clean)) |> kable()
```

	x
region	0

	x
year	0
week	0
total_specimens	2208
total_a	2208
total_b	2208
percent_positive	2208
percent_a	6205
percent_b	6205

- After interpolation and median imputation, a small number of missing values still remained in both datasets.
- In the ILINet dataset, 344 missing values remained in percent_unweighted_ili, ilitotal, num_of_providers, and total_patients.
- In the Clinical Labs dataset, 2,208 missing values still remained in total_specimens, total_a, total_b, and percent_positive, while percent_a and percent_b contained 6,205 missing values.
- Variables percent_a and percent_b were considered redundant because they are derived from existing laboratory variables
- The remaining missing records in clinical laboratory variables will be removed to ensure cleaner and more reliable regression modelling.

```
# Remove remaining missing values from ilinet dataset
ilinet_clean <- ilinet_clean %>%
  drop_na(percent_unweighted_ili,ilitotal,num_of_providers,total_patients)

#Verify remaining missing values in ilinet dataset
colSums(is.na(ilinet_clean)) |> knitr::kable()
```

	x
region	0
year	0
week	0
percent_unweighted_ili	0
ilitotal	0
num_of_providers	0
total_patients	0

```
# Remove redundant variables in Clinical labs dataset
clinical_labs_clean <- clinical_labs_clean %>% select(-percent_a,-percent_b)

#Remaining missing records are removed
clinical_labs_clean <- clinical_labs_clean %>%
  drop_na(total_specimens,total_a,total_b,percent_positive)

# Verify any missing values left in clinical labs dataset
colSums(is.na(clinical_labs_clean)) |> knitr::kable()
```

	x
region	0
year	0
week	0
total_specimens	0
total_a	0
total_b	0
percent_positive	0

Output shows the datasets are clean with no missing values.

MERGING DATASETS

- The ILINet and Clinical Labs datasets were merged using region, year, and week.
- An inner_join() is used to keep only records that existed in both datasets,so each final observation contained both symptom-based influenza activity data and laboratory-confirmed influenza testing information.
- An inner join was chosen because predictive modeling requires complete observations across both datasets, and unmatched records could introduce missing values or inconsistent outbreak information.

```
# Merge datasets using region, year, week.
# INNER JOIN
# Keep only matching influenza records
flu_data <- ilinet_clean %>%
  inner_join(clinical_labs_clean,
    by = c("region","year","week"),
    suffix = c("_ili", "_lab") )

# Verify merged dataset
head(flu_data[,1:8]) |>knitr::kable()
```

region	year	week	percent_unweighted_ili	ilitotal	num_of_providers	total_patients	total_specimens
Alabama	2015	40	2.58875	272	28	10507	167
Alabama	2015	41	2.48819	258	28	10369	167
Alabama	2015	42	2.31635	251	29	10836	193
Alabama	2015	43	2.19702	279	26	12699	164
Alabama	2015	44	2.31715	313	28	13508	254
Alabama	2015	45	2.20919	290	30	13127	250

```
str(flu_data)
```

```
## tibble [27,600 x 11] (S3: tbl_df/tbl/data.frame)
##  $ region                : chr [1:27600] "Alabama" "Alabama" "Alabama" "Alabama" ...
##  $ year                  : int [1:27600] 2015 2015 2015 2015 2015 2015 2015 2015 2015 2015 ...
##  $ week                  : int [1:27600] 40 41 42 43 44 45 46 47 48 49 ...
##  $ percent_unweighted_ili: num [1:27600] 2.59 2.49 2.32 2.2 2.32 ...
##  $ ilitotal              : num [1:27600] 272 258 251 279 313 290 305 255 295 339 ...
##  $ num_of_providers      : num [1:27600] 28 28 29 26 28 30 30 26 27 26 ...
##  $ total_patients        : num [1:27600] 10507 10369 10836 12699 13508 ...
```

```
##  $ total_specimens       : num [1:27600] 167 167 193 164 254 250 279 271 321 277 ...
##  $ total_a               : num [1:27600] 2 2 4 3 2 3 4 0 5 3 ...
##  $ total_b               : num [1:27600] 3 1 1 4 3 3 1 3 4 1 ...
##  $ percent_positive      : num [1:27600] 2.99 1.8 2.59 4.27 1.97 2.4 1.79 1.11 2.8 1.44 ...
```

```
dim(flu_data)
```

```
## [1] 27600    11
```

```
cat("Merged Dataset Records:", nrow(flu_data), "\n")
```

```
## Merged Dataset Records: 27600
```

- The final merged influenza dataset contains 27,600 records and 11 variables, showing that matching observations were successfully combined from both ILINet and Clinical Labs datasets.

```
summary_table <- describe(
  flu_data[, c(
    "percent_positive",
    "percent_unweighted_ili",
    "ilitotal",
    "total_patients",
    "total_specimens",
    "total_a",
    "total_b"
  )]
)

knitr::kable(
  summary_table[, c("n","mean","sd","min","max","skew")],
  digits = 2,
  caption = "Summary Statistics of Key Influenza Variables"
)
```

Table 22: Summary Statistics of Key Influenza Variables

	n	mean	sd	min	max	skew
percent_positive	27600	5.52	8.65	0	66.67	2.03
percent_unweighted_ili	27600	2.14	2.02	0	19.33	2.23
ilitotal	27600	774.17	1625.26	0	26030.00	5.61
total_patients	27600	30258.92	40860.39	0	317409.00	3.40
total_specimens	27600	986.76	1946.21	0	36660.00	6.05
total_a	27600	71.55	322.82	0	9364.00	13.21
total_b	27600	19.34	85.97	0	2054.00	10.32

```
# Exporting the merged dataset as a csv file
write.csv(
  flu_data,
  file = here("merged_dataset.csv"),
  row.names = FALSE
```

```
)

# Variable to count the # of original rows before feature engineering
original_rows <- nrow(flu_data)
```

FEATURE ENGINEERING

Feature engineering was performed to create additional predictors that may help the model better capture patterns in influenza activity. This process included creating future target variables, capturing previous week metrics, rolling averages, changes from week to week, and calculating percentages. Ultimately, the goal of these engineered features are to help improve the model's ability to predict future influenza percent positivity

```
### Creating a new dataset that will contain the engineered features
flu_features <- flu_data %>%
  arrange(region, year, week) %>% # Sorting accordingly to perform rolling calculations
  group_by(region) %>% # Grouping by region to calculate separately per region

  mutate(

    ### 5 columns to predict on
    # Creating 4 other target variables by pulling the leading week's percent positive
    target_1wk = lead(percent_positive, 1),
    target_2wk = lead(percent_positive, 2),
    target_3wk = lead(percent_positive, 3),
    target_4wk = lead(percent_positive, 4),

    ### Previous Week's values using the lag function
    lag1_percent_positive = lag(percent_positive, 1),
    lag1_total_specimens = lag(total_specimens, 1),
    lag1_total_a = lag(total_a, 1),
    lag1_total_b = lag(total_b, 1),
    lag1_ili = lag(percent_unweighted_ili, 1),

    ### Calculating 4 week rolling average - structure is as follows:
    # 1. This means to start at last week's metric
    # 2. Takes the rolling mean(average)
    # 3. This then looks at 3 weeks past the last (4 total)
    # 4. Return N/A if there is not a 4 week history

    # rolling average for total speciman
    roll4_total_specimens = slide_dbl(
      lag(total_specimens),
      mean,
      .before = 3,
      .complete = TRUE
    ),

    # rolling average for total a
    roll4_total_a = slide_dbl(
      lag(total_a),
      mean,
      .before = 3,
```

```
    .complete = TRUE
  ),

  # rolling average for total b
  roll4_total_b = slide_dbl(
    lag(total_b),
    mean,
    .before = 3,
    .complete = TRUE
  ),

  # rolling average for percent unweighted ili
  roll4_ili = slide_dbl(
    lag(percent_unweighted_ili),
    mean,
    .before = 3,
    .complete = TRUE
  ),

  # rolling average for percent positive
  roll4_percent_positive = slide_dbl(
    lag(percent_positive),
    mean,
    .before = 3,
    .complete = TRUE
  ),

  ### Metric Differences
  specimens_change = total_specimens - lag(total_specimens, 1),
  total_a_change = total_a - lag(total_a, 1),
  total_b_change = total_b - lag(total_b, 1),
  ili_change = percent_unweighted_ili - lag(percent_unweighted_ili, 1),
  positive_change = percent_positive - lag(percent_positive, 1),

  ### Percentages
  a_ratio = total_a / total_specimens,
  b_ratio = total_b / total_specimens,
  ili_patient_ratio = ilitotal / total_patients,

  ### Week Featuring
  # As weeks are cylical vs linear, need to show week 52, is closer to 1 than it is to 45
  week_sin = sin(2 * pi * week / 52),
  week_cos = cos(2 * pi * week / 52)
) %>%

# Removes region grouping
ungroup()
```

DATA CLEANING (POST FEATURE ENGINEERING)

After feature engineering, the dataset was evaluated for missing values introduced by the lag features, rolling averages, future target variables, and ratio-based metrics. Missing values associated with the ratio features were further investigated and found to result from observations with zero denominators, producing undefined ratio calculations. Because these observations represented weeks with no recorded specimens or patients, the affected ratio values were imputed with 0. The remaining missing values occurred naturally because observations at the beginning and end of each regional time series did not have sufficient historical or future data to calculate all engineered features. Records containing these remaining missing values were removed to create a complete dataset for predictive modeling. The number and percentage of removed observations were then calculated to assess the impact of the feature engineering process on the final dataset.

```
# Check missing values after feature engineering
tibble(
  Variable = names(flu_features),
  Missing_Values = colSums(is.na(flu_features))
) %>%
  filter(Missing_Values > 0) %>%
  arrange(desc(Missing_Values))
```

```
## # A tibble: 22 x 2
##    Variable               Missing_Values
##    <chr>                           <dbl>
##  1 a_ratio                           552
##  2 b_ratio                           552
##  3 target_4wk                        200
##  4 roll4_total_specimens             200
##  5 roll4_total_a                     200
##  6 roll4_total_b                     200
##  7 roll4_ili                         200
##  8 roll4_percent_positive            200
##  9 target_3wk                        150
## 10 target_2wk                        100
## # i 12 more rows
```

Looking at the table above, many of the missing values can be justified from the fact that they come from the engineered rolling averages and future target columns! Any weeks that don't have the historic or future data will return as null. The fact that they come in multiples of 50 also supports this fact. However, what stood out was the missing values for our ratio columns. We need to explore deeper on why these are showing as null.

```
# Count # of records with 0 total speciman
flu_features %>%
  filter(total_specimens == 0)
```

```
## # A tibble: 552 x 35
##    region       year  week percent_unweighted_ili ilitotal num_of_providers
##    <chr>       <int> <int>                  <dbl>    <dbl>            <dbl>
##  1 Puerto Rico  2015    40                   3.36      156                8
##  2 Puerto Rico  2015    41                   2.97      138                8
##  3 Puerto Rico  2015    42                   4.26      199                9
##  4 Puerto Rico  2015    43                   3.55      179                9
##  5 Puerto Rico  2015    44                   4.99      227                8
##  6 Puerto Rico  2015    45                   3.61      169                8
##  7 Puerto Rico  2015    46                   3.39      156                8
##  8 Puerto Rico  2015    47                   4.12      207                9
##  9 Puerto Rico  2015    48                   4.07      205                9
## 10 Puerto Rico  2015    49                   4.92      235                8
```

```
## # i 542 more rows
## # i 29 more variables: total_patients <dbl>, total_specimens <dbl>,
## #   total_a <dbl>, total_b <dbl>, percent_positive <dbl>, target_1wk <dbl>,
## #   target_2wk <dbl>, target_3wk <dbl>, target_4wk <dbl>,
## #   lag1_percent_positive <dbl>, lag1_total_specimens <dbl>,
## #   lag1_total_a <dbl>, lag1_total_b <dbl>, lag1_ili <dbl>,
## #   roll4_total_specimens <dbl>, roll4_total_a <dbl>, roll4_total_b <dbl>, ...
```

The table above explains the reason for the missing values in the engineered a_ratio and b_ratio variables. In all 552 cases, the denominator of the ratio calculation is equal to zero, resulting in a null value. Because these missing values are caused by division by zero rather than missing data, they were imputed with 0 to avoid unnecessarily removing observations from the dataset

```
flu_features %>%
  filter(is.na(ili_patient_ratio)) %>%
  select(region, year, week, ilitotal, total_patients, ili_patient_ratio)
```

```
## # A tibble: 9 x 6
##   region        year  week ilitotal total_patients ili_patient_ratio
##   <chr>        <int> <int>    <dbl>          <dbl>             <dbl>
## 1 Alaska        2025    39        0              0               NaN
## 2 North Dakota  2023    22        0              0               NaN
## 3 North Dakota  2023    26        0              0               NaN
## 4 Oklahoma      2018    25        0              0               NaN
## 5 Oklahoma      2018    38        0              0               NaN
## 6 South Dakota  2019    22        0              0               NaN
## 7 Utah          2018    34        0              0               NaN
## 8 Utah          2022    26        0              0               NaN
## 9 Virginia      2021    39        0              0               NaN
```

The table above explains the reason for the null values in the engineered ili_patient_ratio variable. Similar to the previous ratio features, the denominator of the calculation is equal to zero, resulting in a null value. This was true for all 9 missing observations. Because these null values were caused by division by zero rather than missing data, they were imputed with 0 to avoid unnecessarily removing observations from the dataset

```
# Imputing engineered ratio columns with 0
flu_features <- flu_features %>%
  mutate(
    a_ratio = replace_na(a_ratio, 0),
    b_ratio = replace_na(b_ratio, 0),
    ili_patient_ratio = replace_na(ili_patient_ratio, 0)
  )

# Check missing values after imputation
tibble(
  Variable = names(flu_features),
  Missing_Values = colSums(is.na(flu_features))
) %>%
  filter(Missing_Values > 0) %>%
  arrange(desc(Missing_Values))
```

```
## # A tibble: 19 x 2
```

```
##    Variable               Missing_Values
##    <chr>                           <dbl>
##  1 target_4wk                        200
##  2 roll4_total_specimens             200
##  3 roll4_total_a                     200
##  4 roll4_total_b                     200
##  5 roll4_ili                         200
##  6 roll4_percent_positive            200
##  7 target_3wk                        150
##  8 target_2wk                        100
##  9 target_1wk                         50
## 10 lag1_percent_positive              50
## 11 lag1_total_specimens               50
## 12 lag1_total_a                       50
## 13 lag1_total_b                       50
## 14 lag1_ili                           50
## 15 specimens_change                   50
## 16 total_a_change                     50
## 17 total_b_change                     50
## 18 ili_change                         50
## 19 positive_change                    50
```

```r
# Remove rows with missing values
flu_model_data <- flu_features %>%
  drop_na()

# Count removed rows
final_rows <- nrow(flu_model_data)
rows_removed <- original_rows - final_rows

# Displaying amount of rows removed from feature engineering
cat("Original rows:", original_rows, "\n")
```

```
## Original rows: 27600
```

```r
cat("Rows after feature engineering cleanup:", final_rows, "\n")
```

```
## Rows after feature engineering cleanup: 27200
```

```r
cat("Rows removed:", rows_removed, "\n")
```

```
## Rows removed: 400
```

```r
cat("Percent removed:", round(rows_removed / original_rows * 100, 2), "%\n")
```

```
## Percent removed: 1.45 %
```

EXPLORING ENGINEERED DATASET

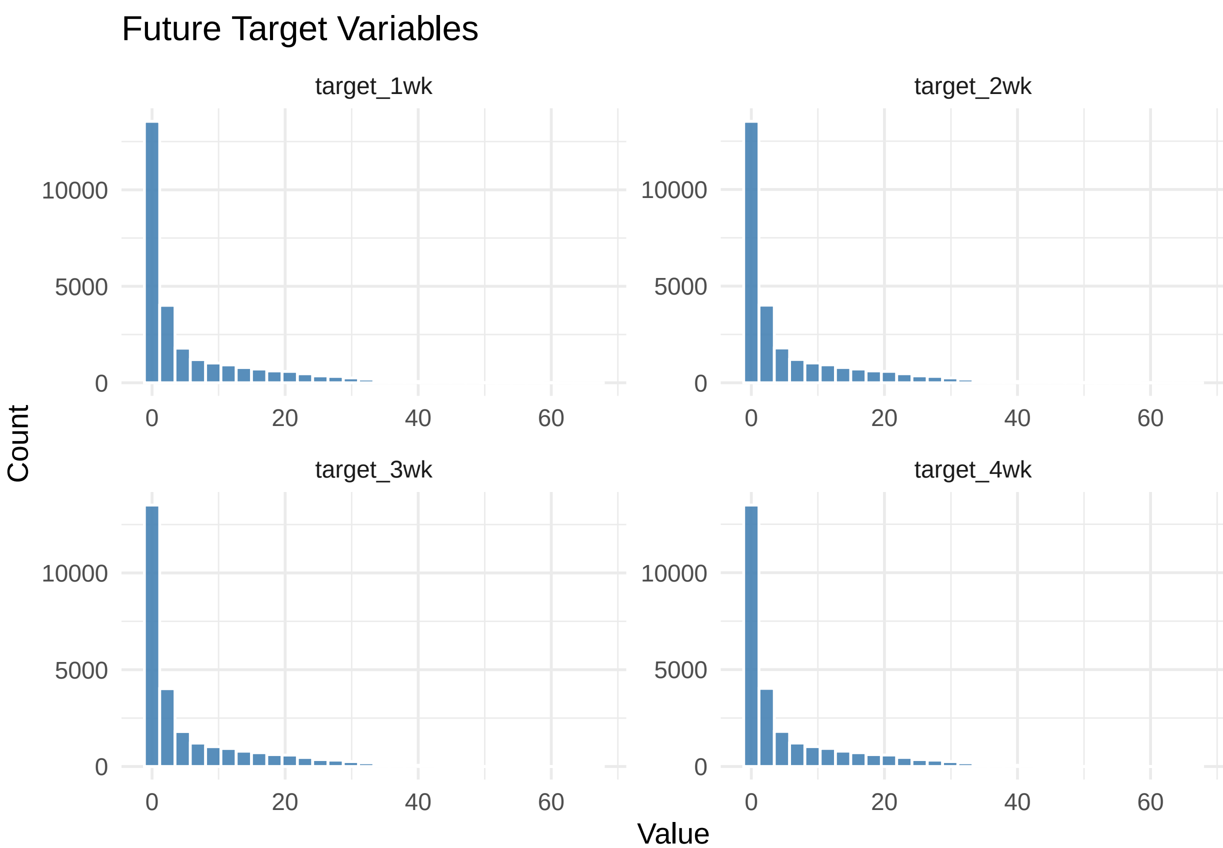
```r
# Viewing our engineered dataset
str(flu_model_data)
```

```
## tibble [27,200 x 35] (S3: tbl_df/tbl/data.frame)
##  $ region                 : chr [1:27200] "Alabama" "Alabama" "Alabama" "Alabama" ...
##  $ year                   : int [1:27200] 2015 2015 2015 2015 2015 2015 2015 2015 2015 2016 ...
##  $ week                   : int [1:27200] 44 45 46 47 48 49 50 51 52 1 ...
##  $ percent_unweighted_ili: num [1:27200] 2.32 2.21 2.65 3.5 2.45 ...
##  $ ilitotal               : num [1:27200] 313 290 305 255 295 339 334 249 220 293 ...
##  $ num_of_providers       : num [1:27200] 28 30 30 26 27 26 25 26 23 22 ...
##  $ total_patients         : num [1:27200] 13508 13127 11531 7279 12023 ...
##  $ total_specimens        : num [1:27200] 254 250 279 271 321 277 250 257 287 262 ...
##  $ total_a                : num [1:27200] 2 3 4 0 5 3 3 2 2 3 ...
##  $ total_b                : num [1:27200] 3 3 1 3 4 1 6 1 3 2 ...
##  $ percent_positive       : num [1:27200] 1.97 2.4 1.79 1.11 2.8 1.44 3.6 1.17 1.74 1.91 ...
##  $ target_1wk             : num [1:27200] 2.4 1.79 1.11 2.8 1.44 3.6 1.17 1.74 1.91 2.94 ...
##  $ target_2wk             : num [1:27200] 1.79 1.11 2.8 1.44 3.6 1.17 1.74 1.91 2.94 3.11 ...
##  $ target_3wk             : num [1:27200] 1.11 2.8 1.44 3.6 1.17 1.74 1.91 2.94 3.11 1.87 ...
##  $ target_4wk             : num [1:27200] 2.8 1.44 3.6 1.17 1.74 1.91 2.94 3.11 1.87 2.38 ...
##  $ lag1_percent_positive : num [1:27200] 4.27 1.97 2.4 1.79 1.11 2.8 1.44 3.6 1.17 1.74 ...
##  $ lag1_total_specimens  : num [1:27200] 164 254 250 279 271 321 277 250 257 287 ...
##  $ lag1_total_a          : num [1:27200] 3 2 3 4 0 5 3 3 2 2 ...
##  $ lag1_total_b          : num [1:27200] 4 3 3 1 3 4 1 6 1 3 ...
##  $ lag1_ili              : num [1:27200] 2.2 2.32 2.21 2.65 3.5 ...
##  $ roll4_total_specimens : num [1:27200] 173 194 215 237 264 ...
##  $ roll4_total_a         : num [1:27200] 2.75 2.75 3 3 2.25 3 3 2.75 3.25 2.5 ...
##  $ roll4_total_b         : num [1:27200] 2.25 2.25 2.75 2.75 2.5 2.75 2.25 3.5 3 2.75 ...
##  $ roll4_ili             : num [1:27200] 2.4 2.33 2.26 2.34 2.67 ...
##  $ roll4_percent_positive: num [1:27200] 2.91 2.66 2.81 2.61 1.82 ...
##  $ specimens_change      : num [1:27200] 90 -4 29 -8 50 -44 -27 7 30 -25 ...
##  $ total_a_change        : num [1:27200] -1 1 1 -4 5 -2 0 -1 0 1 ...
##  $ total_b_change        : num [1:27200] -1 0 -2 2 1 -3 5 -5 2 -1 ...
##  $ ili_change            : num [1:27200] 0.12 -0.108 0.436 0.858 -1.05 ...
##  $ positive_change       : num [1:27200] -2.3 0.43 -0.61 -0.68 1.69 -1.36 2.16 -2.43 0.57 0.17 ...
##  $ a_ratio               : num [1:27200] 0.00787 0.012 0.01434 0 0.01558 ...
##  $ b_ratio               : num [1:27200] 0.01181 0.012 0.00358 0.01107 0.01246 ...
##  $ ili_patient_ratio     : num [1:27200] 0.0232 0.0221 0.0265 0.035 0.0245 ...
##  $ week_sin              : num [1:27200] -0.823 -0.749 -0.663 -0.568 -0.465 ...
##  $ week_cos              : num [1:27200] 0.568 0.663 0.749 0.823 0.885 ...
```

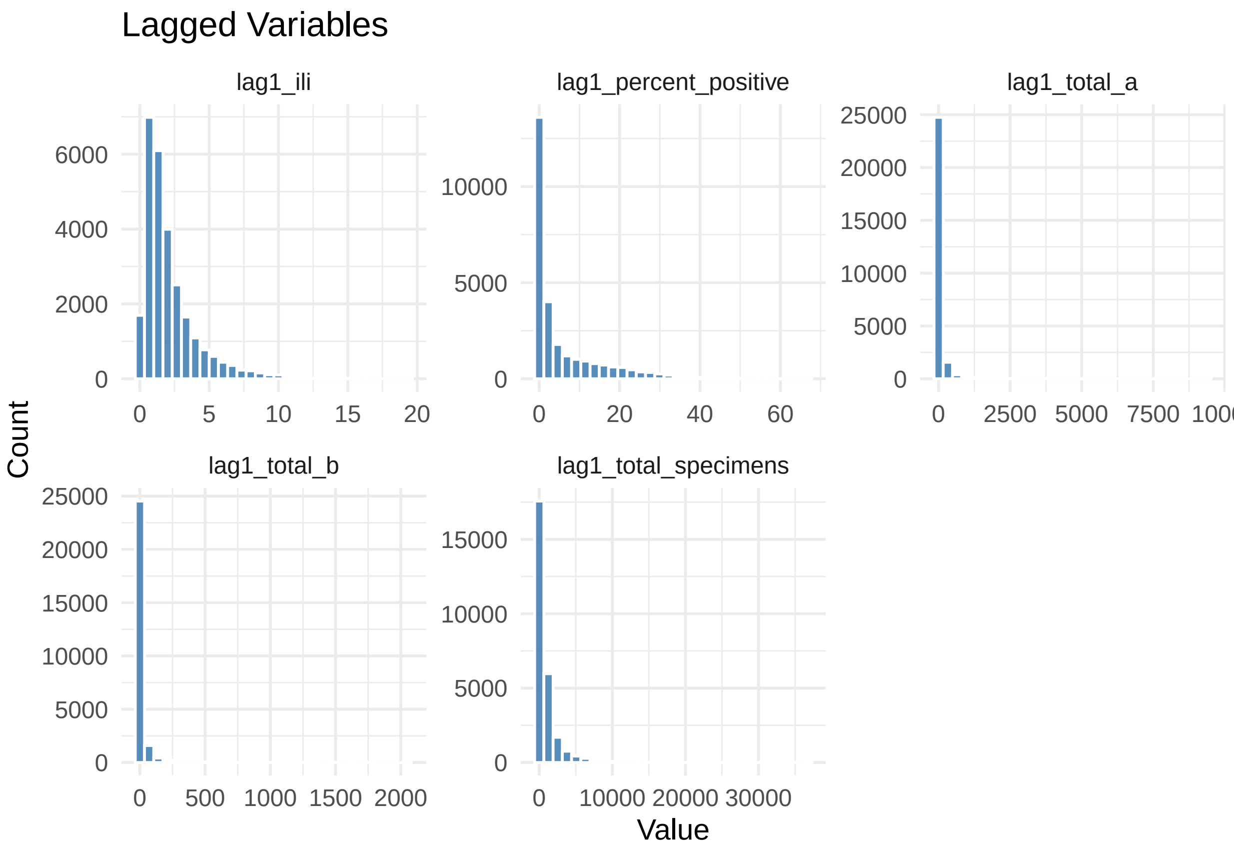
ENGINEERED FEATURE DISTRIBUTIONS

```r
# Future target variables
future_targets <- flu_model_data %>%
  select(starts_with("target"))

#Convert all numeric columns into one common structure(long format)so ggplot create multiple panels
future_targets_long <- future_targets %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Value")

#Create histogram distributions for all numeric variables
```

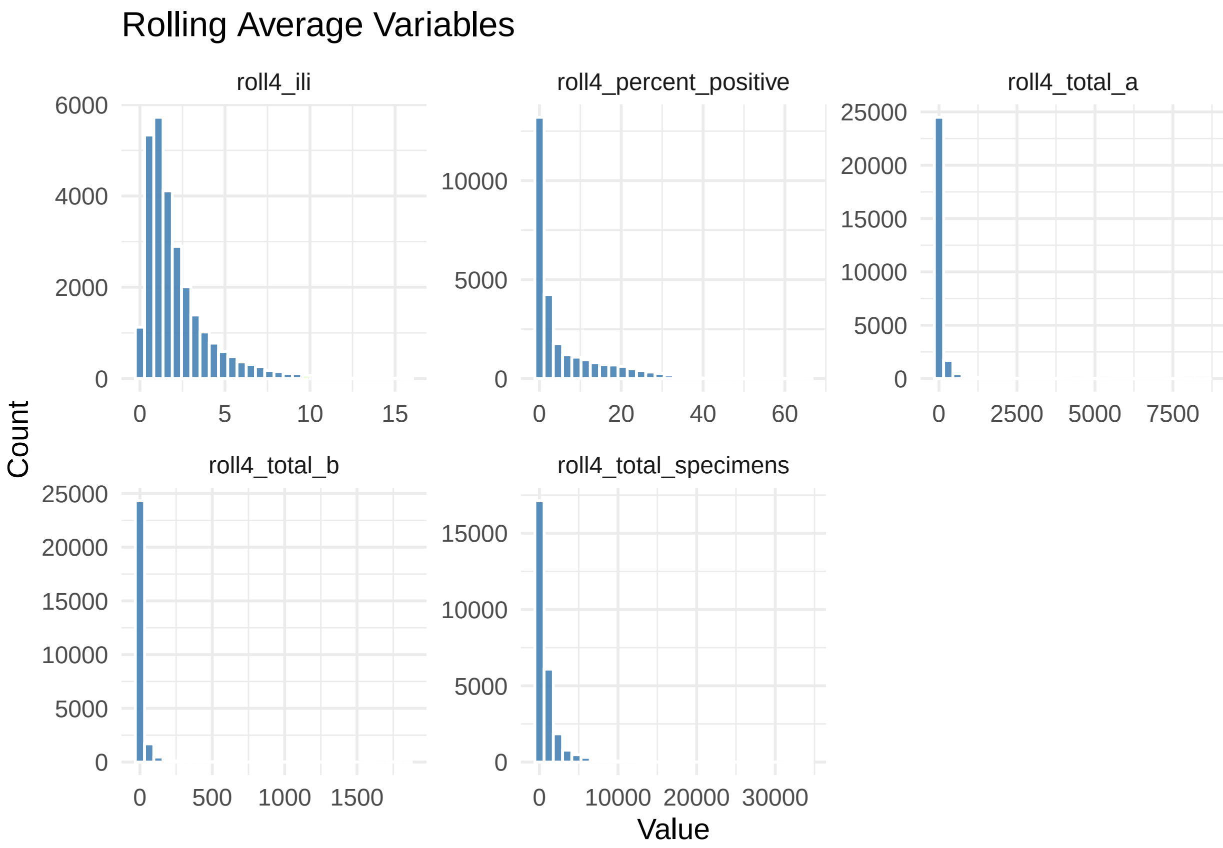
```r
ggplot(future_targets_long, aes(x = Value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white", alpha = 0.9) +
  facet_wrap(~ Variable, scales = "free", ncol = 2) +
  labs(title = "Future Target Variables",
       x = "Value", y = "Count") +
  theme_minimal()
```



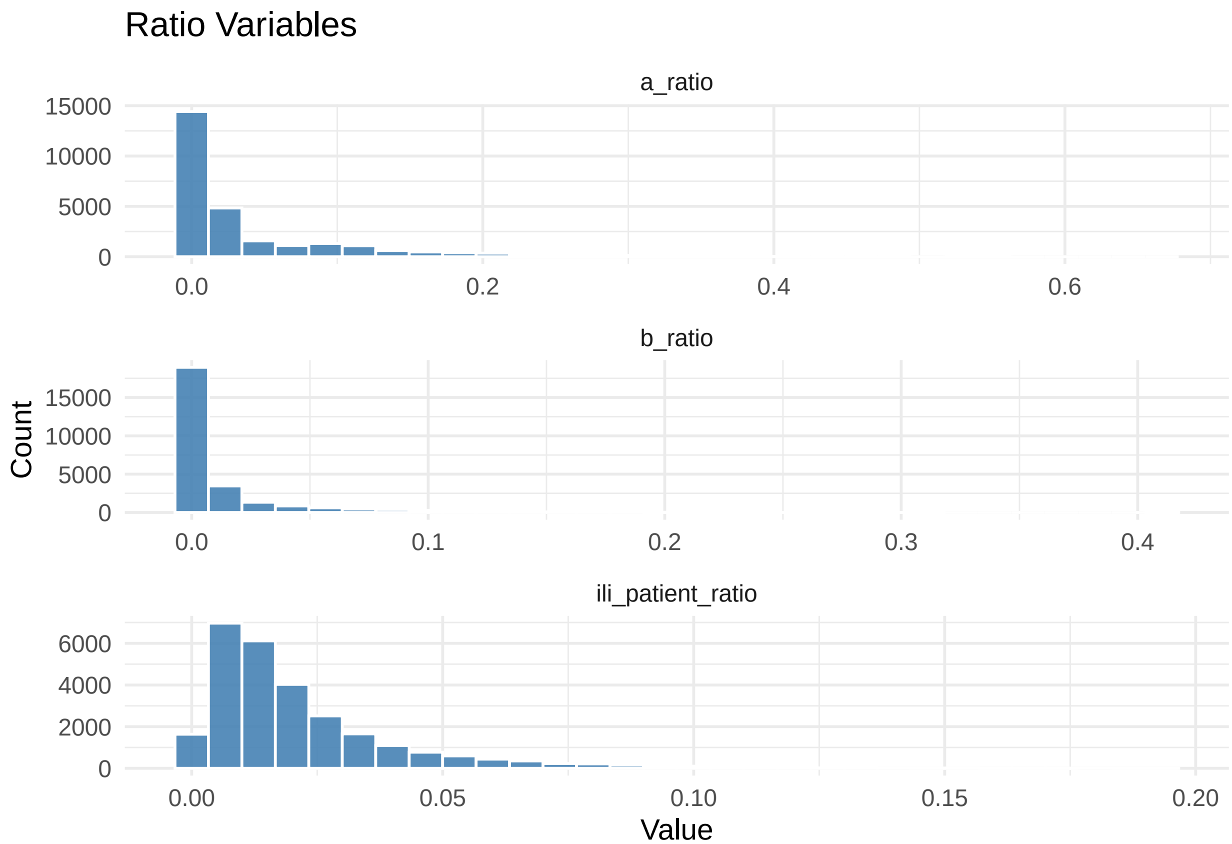

```r
# Lagged variables
lagged_vars <- flu_model_data %>%
  select(starts_with("lag"))

#Convert all numeric columns into one common structure(long format)so ggplot can create multiple panels
lagged_vars_long <- lagged_vars %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Value")

# Create histogram distributions for all lagged variables
ggplot(lagged_vars_long, aes(x = Value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white", alpha = 0.9) +
  facet_wrap(~ Variable, scales = "free", ncol = 3) +
  labs(title = "Lagged Variables",
       x = "Value", y = "Count") +
  theme_minimal()
```




```r
# Rolling average variables
rolling_vars <- flu_model_data %>%
  select(starts_with("roll"))

#Convert all numeric columns into one common structure(long format)so ggplot can create multiple panels
rolling_vars_long <- rolling_vars %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Value")

# Create histogram distributions for all rolling average variables
ggplot(rolling_vars_long, aes(x = Value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white", alpha = 0.9) +
  facet_wrap(~ Variable, scales = "free", ncol = 3) +
  labs(title = "Rolling Average Variables",
       x = "Value", y = "Count") +
  theme_minimal()
```




```
# Ratio variables
ratio_vars <- flu_model_data %>%
  select(contains("ratio"))

#Convert all numeric columns into one common structure (long format)so ggplot can create multiple panel
ratio_vars_long <- ratio_vars %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Value")

# Create histogram distributions for all ratio variables
ggplot(ratio_vars_long, aes(x = Value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white", alpha = 0.9) +
  facet_wrap(~ Variable, scales = "free", ncol = 1) +
  labs(title = "Ratio Variables",
       x = "Value", y = "Count") +
  theme_minimal()
```



As we can see, many of the engineered variables remain highly right-skewed. This may pose challenges for certain modeling techniques that are sensitive to predictor distributions. However, several engineered variables, contain negative values, making a Box-Cox transformation unsuitable since it requires strictly positive data.

FEATURE TRANSFORMATION

Log Transformation

```
## Log transformation for highly skewed count variables
# Variables transformed using log1p()
vars <- c("ilitotal","total_patients","total_specimens","total_a","total_b",
  "lag1_total_specimens","lag1_total_a","lag1_total_b","roll4_total_specimens",
  "roll4_total_a","roll4_total_b")

# Create log-transformed variables
for(v in vars) {
  flu_model_data[[paste0("log_", v)]] <- log1p(flu_model_data[[v]])
}

# Compare skewness before and after transformation
skew_comparison <- data.frame(
  Variable = vars,
  Before = sapply(
    vars,
    function(x) skewness(flu_model_data[[x]], na.rm = TRUE)
```

```r
  ),
  After = sapply(
    paste0("log_", vars),
    function(x) skewness(flu_model_data[[x]], na.rm = TRUE)
  ),
  row.names = NULL
)
knitr::kable( skew_comparison,digits = 3,
  caption = "Skewness Before and After Log Transformation")
```

Table 23: Skewness Before and After Log Transformation

Variable	Before	After
ilitotal	5.595	-0.492
total_patients	3.404	-0.402
total_specimens	6.040	-0.663
total_a	13.116	0.884
total_b	10.301	1.443
lag1_total_specimens	6.049	-0.662
lag1_total_a	13.116	0.886
lag1_total_b	10.353	1.448
roll4_total_specimens	5.940	-0.675
roll4_total_a	12.369	0.885
roll4_total_b	9.944	1.454

- The most highly skewed variables included lag1_total_a (13.12), roll4_total_a (12.37), lag1_total_b (10.35), roll4_total_b (9.94), lag1_total_specimens (6.05), and roll4_total_specimens (5.94).
- To address this issue, a log transformation using log1p() was applied to the original count variables (ilitotal, total_patients, total_specimens, total_a, total_b) as well as the lagged and rolling average count variables (lag1_total_specimens, lag1_total_a, lag1_total_b, roll4_total_specimens, roll4_total_a, and roll4_total_b).
- The transformation helped reduce the impact of extreme values, improve data distribution, and support better regression model performance.

Categorical Variable transformation

```r
#Categorical transformation
flu_model_data$region <- as.factor(flu_model_data$region)
```

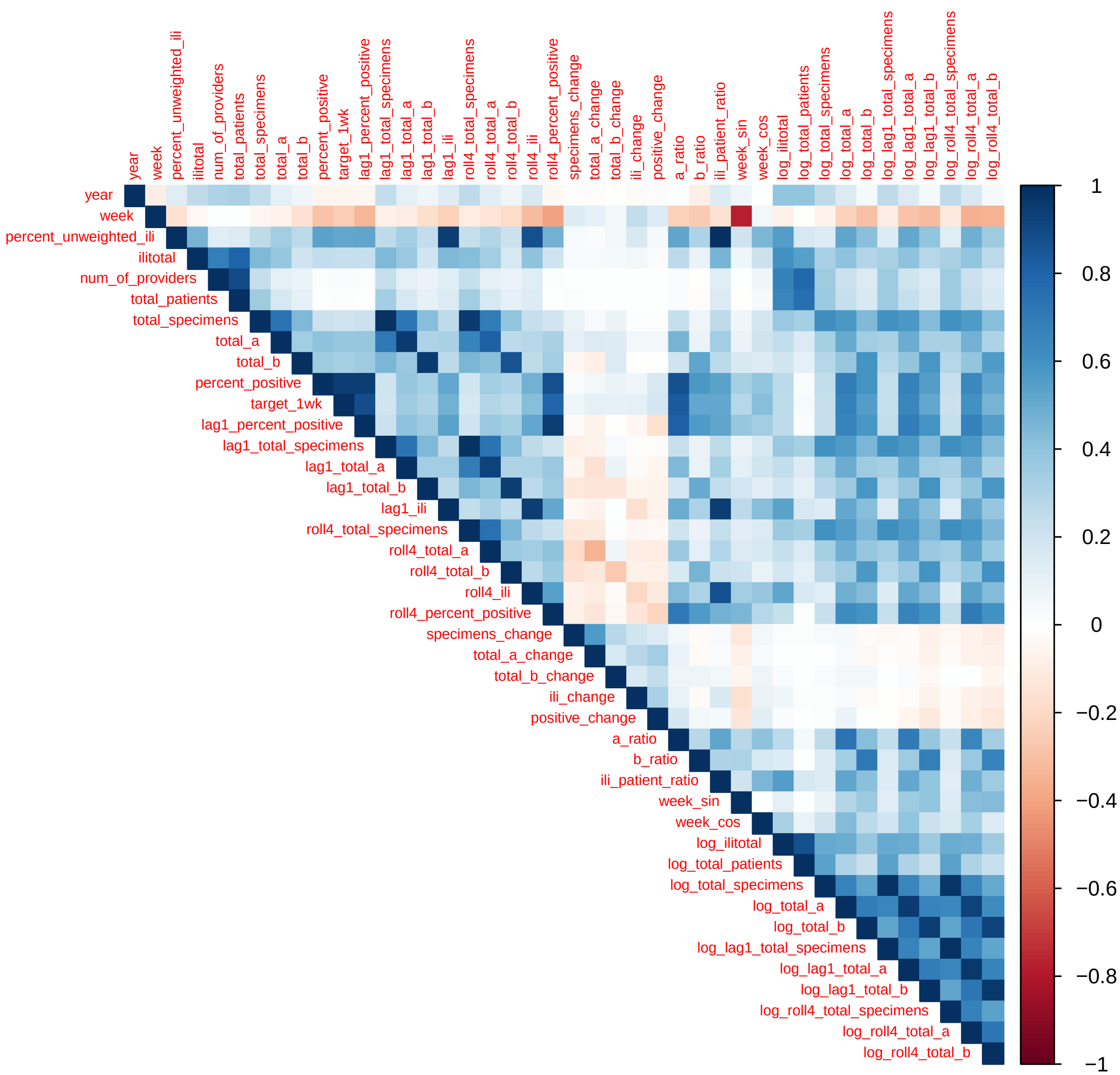
The region variable was converted to a categorical factor to represent geographic locations appropriately during modeling.

EXPLORATORY DATA ANALYSIS

Correlation Heatmap

Which variables are most strongly associated with 1-week-ahead influenza positivity (target_1wk)?

```r
#Select only numeric variables for correlation analysis
numeric_data <- flu_model_data %>% select(where(is.numeric)) %>%
  select(-target_2wk,-target_3wk,-target_4wk)

#Compute correlation matrix
cor_matrix <- cor(numeric_data,use = "complete.obs")

#plot correlation heatmap
corrplot(cor_matrix,method = "color",type = "upper",tl.cex = 0.6)
```




```r
# Correlation of all variables with the target variable
target_corr <- cor_matrix[, "target_1wk"]

target_corr_df <- tibble(Variable = names(target_corr),Correlation = unname(target_corr))
```

```r
#sort in ascending order
target_corr_df <- target_corr_df %>%
  filter(Variable != "target_1wk") %>%
  arrange(desc(abs(Correlation)))
kable(
  head(target_corr_df, 10),digits = 3,caption = "Top Variables Correlated with Target Variable")
```

Table 24: Top Variables Correlated with Target Variable

Variable	Correlation
percent_positive	0.942
lag1_percent_positive	0.880
a_ratio	0.830
roll4_percent_positive	0.792
log_total_a	0.678
log_lag1_total_a	0.653
log_roll4_total_a	0.608
log_total_b	0.552
b_ratio	0.520
percent_unweighted_ili	0.517

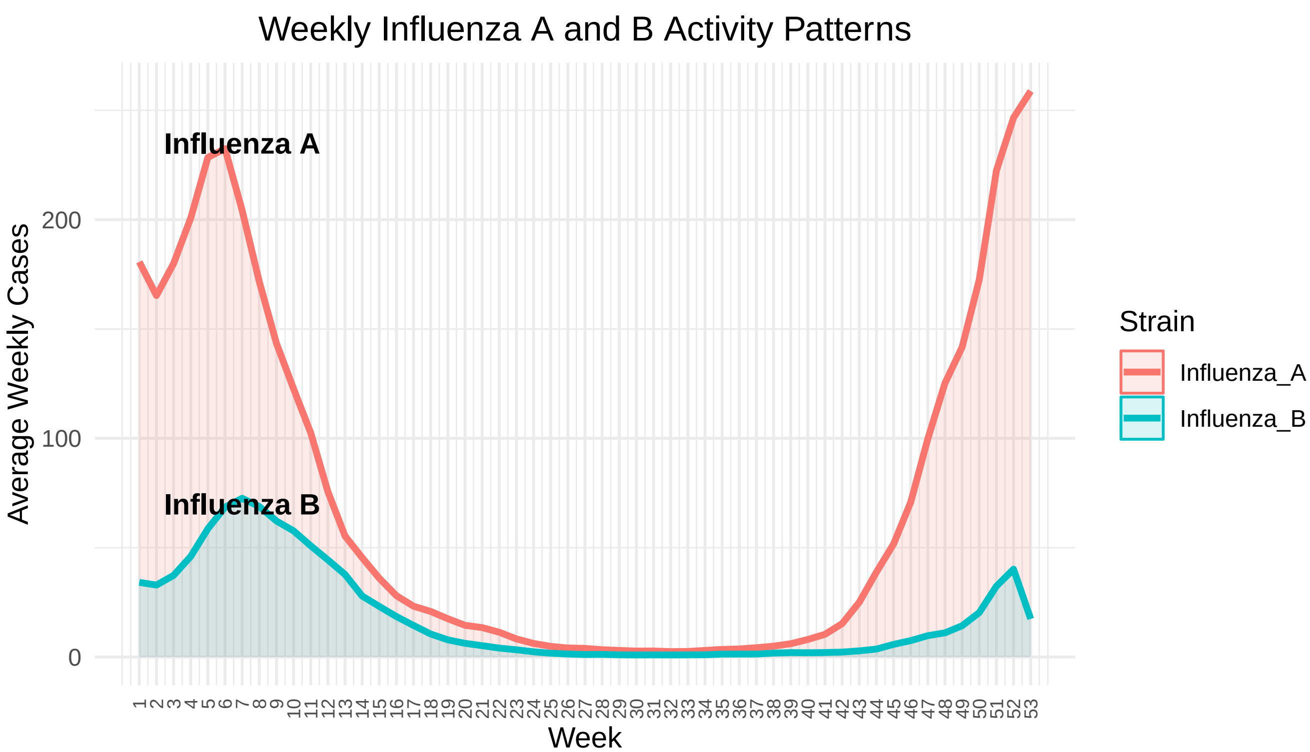
- The strongest correlations with target were lag1_percent_positive (0.942), a_ratio (0.877), and roll4_percent_positive (0.871). This indicates that previous influenza activity and recent outbreak trends are strongly related to current influenza positivity.
- Laboratory testing variables also showed positive correlations with influenza positivity, log_total_a (0.697), log_lag1_total_a (0.678), log_roll4_total_a (0.646), log_total_b (0.590), and log_lag1_total_b (0.553). This suggests that influenza-positive laboratory specimens are useful indicators of outbreak severity.
- percent_unweighted_ili (0.539) showed a moderate positive correlation with percent_positive,showing that higher influenza-like illness activity is associated with higher laboratory-confirmed influenza positivity.
- The strong correlations of lag1_percent_positive, roll4_percent_positive, and laboratory testing variables support the use of historical surveillance data and engineered features for predicting future influenza activity.

Influenza Strain Contribution

Which influenza strain contributes most to outbreaks?

```r
ab_weekly <- flu_model_data %>% group_by(week) %>%
  summarise(Influenza_A = mean(total_a, na.rm = TRUE),
    Influenza_B = mean(total_b, na.rm = TRUE)) %>%
  pivot_longer(cols = c(Influenza_A, Influenza_B),
    names_to = "strain",values_to = "cases")

# Labels placed near the peaks
line_labels <- data.frame(
  week = c(7, 7),cases = c(235, 70),label = c("Influenza A", "Influenza B"))

ggplot(ab_weekly,
  aes(x = week,y = cases,color = strain,fill = strain )) +
```

```r
  geom_area(alpha = 0.15,position = "identity") +geom_line(linewidth = 1.2 ) +
  geom_text(data = line_labels,aes(x = week, y = cases, label = label),
    inherit.aes = FALSE,fontface = "bold",size = 4) +
  scale_x_continuous(breaks = 1:53,limits = c(1, 53)) +
  labs(title = "Weekly Influenza A and B Activity Patterns",
    x = "Week",y = "Average Weekly Cases",color = "Strain",fill = "Strain") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
    axis.text.x = element_text(angle = 90,size = 7,vjust = 0.5,hjust = 1))
```



- The chart compares Influenza A and Influenza B activity during the peak outbreak week of each year.
- This suggests that Influenza A is the primary driver of annual influenza outbreak peaks.

Regional Influenza Activity

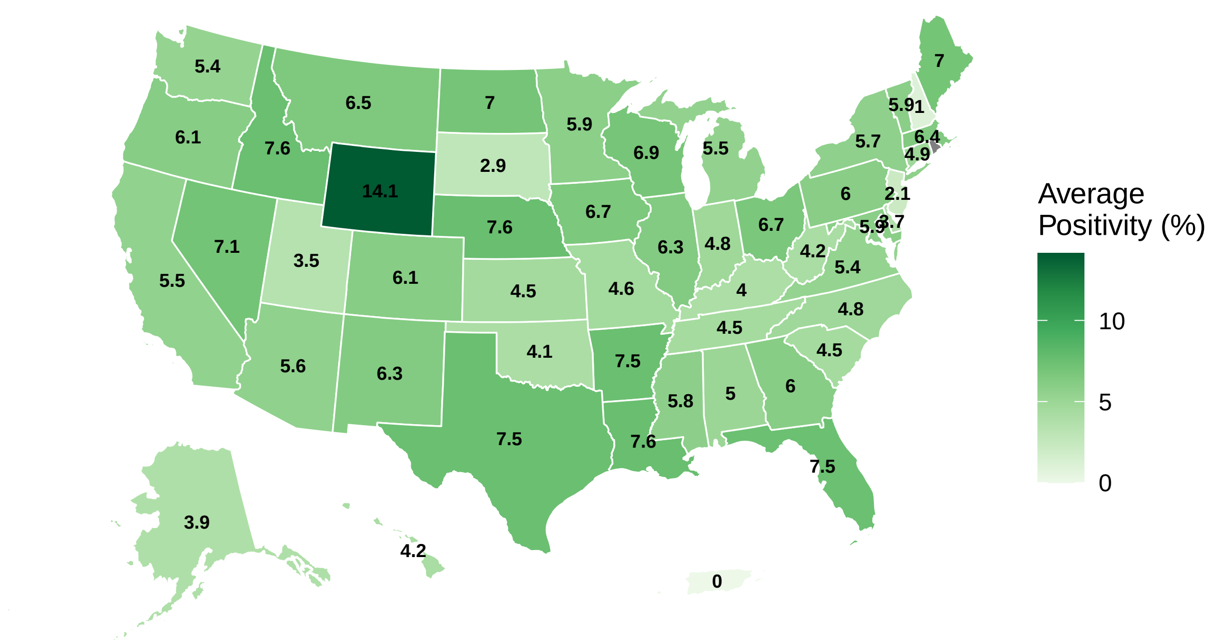
Which regions have experienced the highest influenza activity during the past years and may require additional healthcare preparedness?

```r
# Average influenza positivity by state
map_data <- flu_model_data %>% group_by(region) %>%
  summarise(avg_positive = mean(percent_positive, na.rm = TRUE),
    .groups = "drop") %>% rename(full = region)

# US state map
states_sf <- usmap::us_map(regions = "states") %>% st_as_sf()

# Join influenza data
```

```r
label_data <- states_sf %>% left_join(map_data, by = "full")

# Create state centroids for labels
centroids <- st_centroid(label_data)
coords <- st_coordinates(centroids)

label_data$x <- coords[,1]
label_data$y <- coords[,2]

# Map
ggplot(label_data) + geom_sf(aes(fill = avg_positive),color = "white",linewidth = 0.3) +

  geom_text(aes(x = x,y = y,
      label = round(avg_positive, 1)),color = "black",size = 2.5,fontface = "bold") +

  scale_fill_distiller(palette = "violet",direction = 1,name = "Average\nPositivity (%)" ) +

  labs(title = "Average Influenza Positivity by State",
    subtitle = "For All Available Years",
    caption = "Higher values indicate greater influenza activity") +
  theme_void() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 14),
    plot.subtitle = element_text(hjust = 0.5, size = 11),
    legend.position = "right",plot.caption = element_text(hjust = 0.5))
```

Average Influenza Positivity by State

For All Available Years



Higher values indicate greater influenza activity

```r
# Display Top 10 states with highest influenza positivity
top_states <- map_data %>% arrange(desc(avg_positive)) %>% slice_head(n = 10)

knitr::kable(top_states,digits = 2,caption = "Top 10 States with Highest Influenza Activity")
```

Table 25: Top 10 States with Highest Influenza Activity

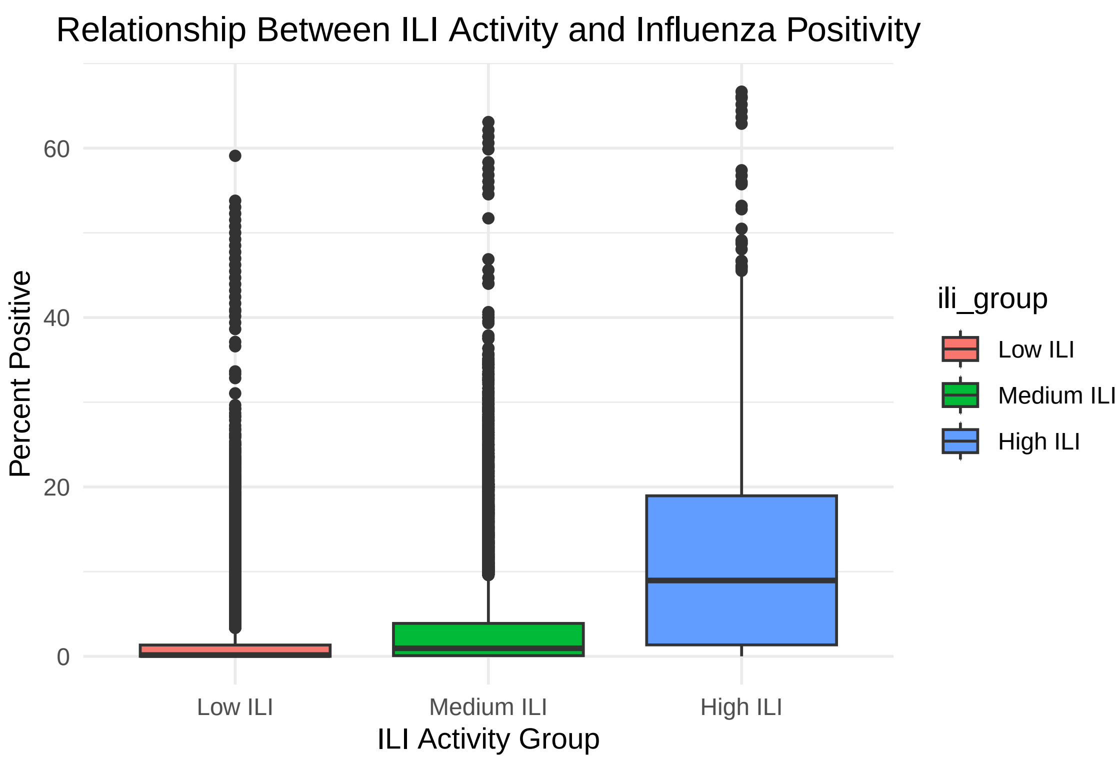
full	avg_positive
Wyoming	14.14
Louisiana	7.64
Nebraska	7.61
Idaho	7.55
Texas	7.53
Arkansas	7.52
Florida	7.47
Nevada	7.11
North Dakota	7.03
Maine	7.00

- The map shows that influenza activity varies across states.
- Wyoming had the highest average influenza positivity, followed by Louisiana, Nebraska, Idaho, and Texas.
- States with higher positivity rates may face greater healthcare demand during influenza seasons.
- The geographic variation suggests that influenza activity is not evenly distributed across the United States.

ILI Activity vs Influenza Positivity

Do regions with more flu-like illness (ILI) also have higher laboratory-confirmed influenza positivity?

```r
#divides all observations into three groups based on percent_unweighted_ili
# Create Low, Medium, and High ILI groups
flu_model_data <- flu_model_data %>%
  mutate(ili_group = cut( percent_unweighted_ili, breaks = quantile(percent_unweighted_ili,
        probs = c(0, 0.33, 0.67, 1),na.rm = TRUE),
      include.lowest = TRUE,labels = c("Low ILI","Medium ILI","High ILI")))

#Create boxplot to compare influenza positivity
  ggplot(flu_model_data,
  aes(ili_group,percent_positive,fill = ili_group)) + geom_boxplot() +theme_minimal() +
    labs(
    title = "Relationship Between ILI Activity and Influenza Positivity",
    x = "ILI Activity Group",y = "Percent Positive") +
     theme(legend.position = "right",plot.title = element_text(hjust = 0.5))
```



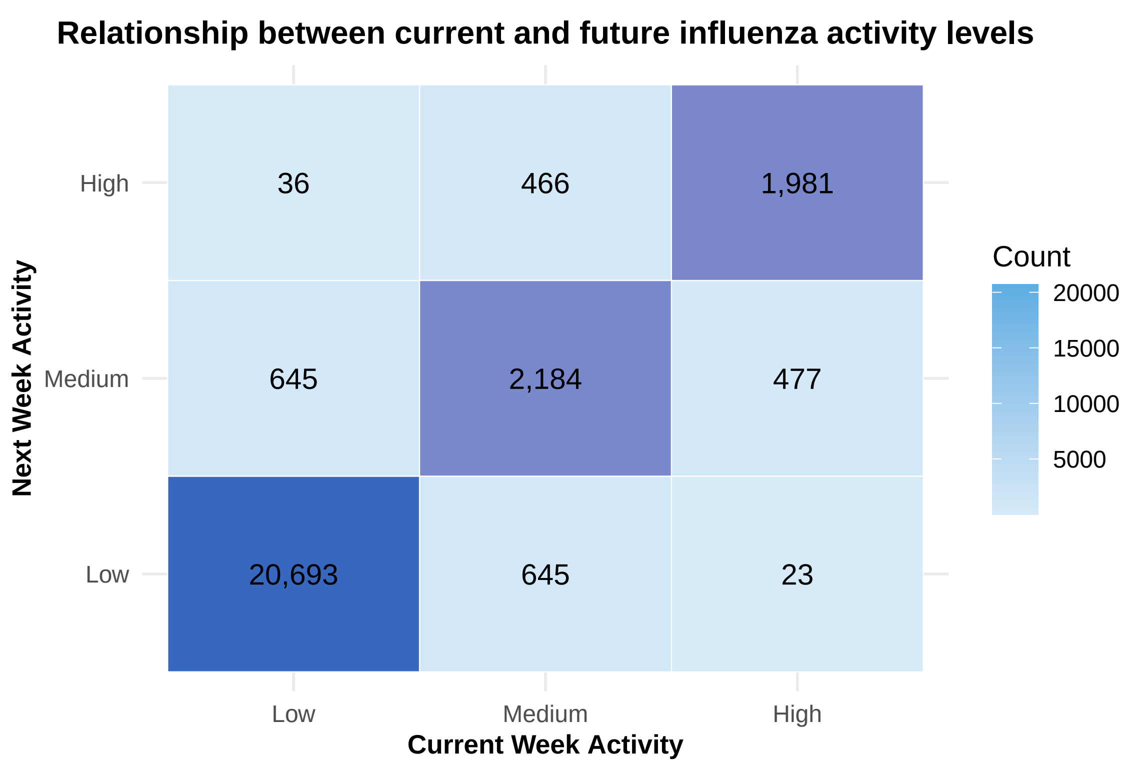
- Regions were grouped into Low, Medium, and High flu-like illness (ILI) activity categories based on percent_unweighted_ili.
- The boxplot shows influenza positivity for regions reporting flu-like symptoms with Low, Medium, and High ILI activity.
- The larger spread in the High ILI group indicates greater variation in influenza positivity during periods of increased flu-like illness activity.
- Regions with High ILI activity generally have higher influenza positivity than regions with Low or Medium ILI activity.
- This means that when more people report flu-like symptoms, there are usually more laboratory-confirmed influenza cases.
- The results show a positive relationship between ILI activity and influenza positivity meaning percent_unweighted_ili is likely to be an important predictor for forecasting future influenza outbreaks.

Influenza Persistence Analysis

Does current influenza activity help predict next week's activity?

```r
# Create influenza activity categories based on percent positive
flu_persistence <- flu_model_data %>%
  arrange(region, year, week) %>%      # Keep data in time order
  group_by(region) %>%                 # Calculate within each region
  mutate(current_level = case_when(percent_positive < 10 ~ "Low",
                                   percent_positive < 20 ~ "Medium",TRUE ~ "High"),

    # Next week's influenza activity
    next_level = lead(current_level)) %>% ungroup()

# Count combinations of current and next week activity
```

```r
heat_data <- flu_persistence %>% filter(!is.na(next_level)) %>%
  count(current_level, next_level)

# Count combinations of current and next week activity levels
heat_data$current_level <- factor(heat_data$current_level,levels = c("Low","Medium","High"))

heat_data$next_level <- factor(heat_data$next_level,levels = c("Low","Medium","High"))

# Create heatmap
heat_data$diag <- heat_data$current_level == heat_data$next_level

ggplot(heat_data, aes(x = current_level, y = next_level)) +

  # Base heatmap
  geom_tile(aes(fill = n), color = "white") +

  # Darken diagonal cells
  geom_tile(data = subset(heat_data, diag),fill = "darkblue",alpha = 0.4,color = "white") +
  geom_text(aes(label = scales::comma(n)), size = 4) +
  scale_fill_gradient(low = "#D6EAF8",high = "#5DADE2",name = "Count") +
  labs(title = "Relationship between current and future influenza activity levels",
    x = "Current Week Activity", y = "Next Week Activity") +
  theme_minimal() + theme(plot.title = element_text(face = "bold",size=12),
  axis.title = element_text(face = "bold",size = 10))+ theme(legend.position = "right",
                                                   plot.title = element_text(hjust = 0.5))
```



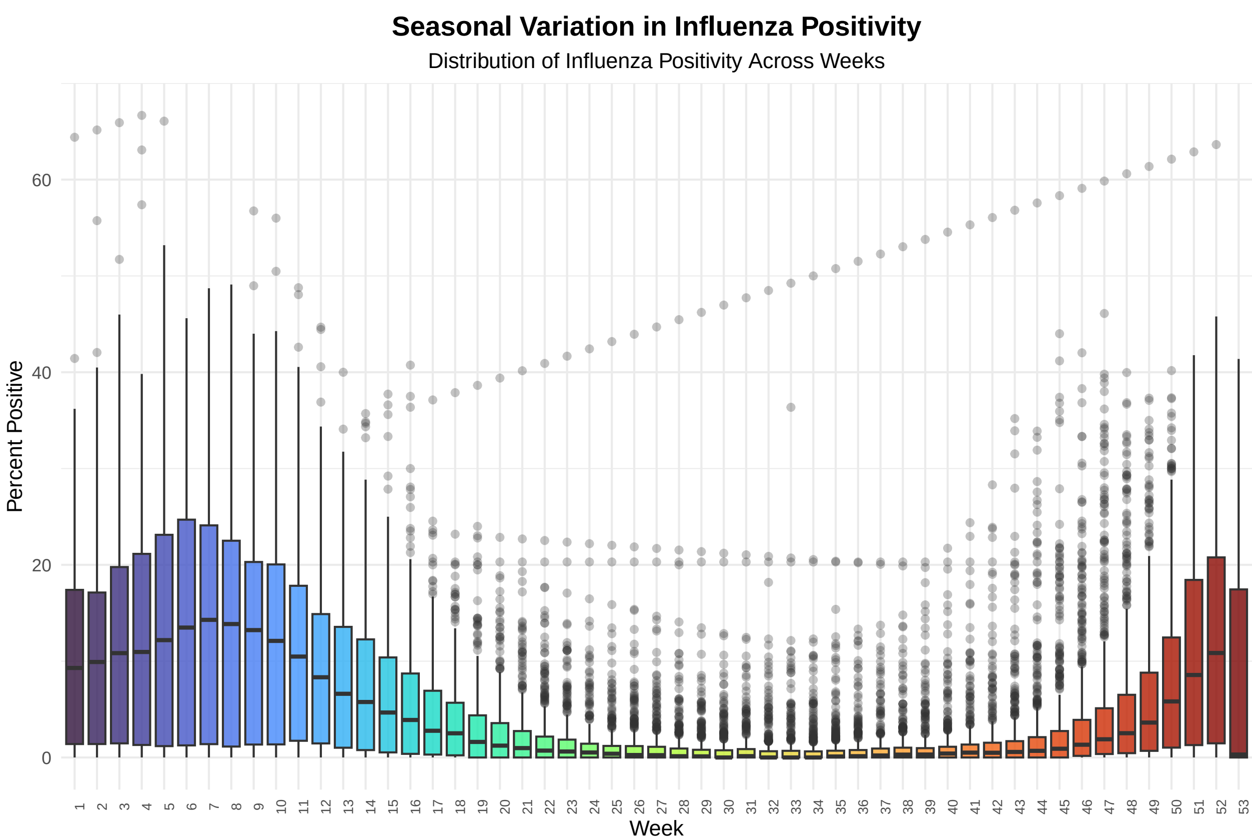
- There were 20,693 weeks where influenza activity was Low this week and also Low the following week.
- There were 1,981 weeks where influenza activity was High this week and remained High next week.
- There were 645 weeks where influenza activity was Low this week but increased to Medium next week.

- Influenza activity generally changes gradually rather than suddenly.
- This indicates that current influenza activity provides useful information about future influenza activity and supports the use of historical influenza measures as predictors in the forecasting model.

Seasonal Boxplot by Week

How does influenza positivity vary throughout the year?

```
# Each box shows the distribution of influenza positivity
# across all years for that specific week
ggplot(flu_model_data,
  aes(factor(week),percent_positive,fill = factor(week) )) +
  geom_boxplot( alpha = 0.8,outlier.alpha = 0.3) + # Display distribution, median,and outliers
  scale_fill_viridis_d(option = "turbo" ) +
  labs(title = "Seasonal Variation in Influenza Positivity",
    subtitle = "Distribution of Influenza Positivity Across Weeks",
    x = "Week", y = "Percent Positive") +theme_minimal() +
  theme(legend.position = "none",plot.title = element_text(face = "bold",size = 14),
    axis.text.x = element_text(angle = 90,size = 7 ))+
  theme(plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5) )
```



- The boxplot shows how influenza positivity changes throughout the year.
- Each box represents one calendar week and summarizes influenza positivity across all years in the dataset.

- Higher positivity levels are observed during the beginning and end of the year, indicating peak influenza seasons.
- Influenza positivity is generally low during the middle weeks of the year.
- Several weeks show large variations and extreme values, meaning that influenza outbreaks can vary in severity from year to year.
- The results confirm that influenza activity follows a strong seasonal pattern.

Modeling

Feature Selection

```
# Predictors to remove: total_a, total_b, percent_positive,percent_unweighted_ili, ilitotal,
# total_a, total_b
cols_to_remove <- c("total_a", "total_b", "percent_positive", "percent_unweighted_ili", "ilitotal")
flu_model_data1 <- flu_model_data %>% dplyr::select(-any_of(cols_to_remove))
#str(flu_model_data1)
```

The original predictors from which features were engineered were removed given their high correlation with the engineered features. The decision to include or exclude additional variables will be determined based on variable importance after training a baseline model.

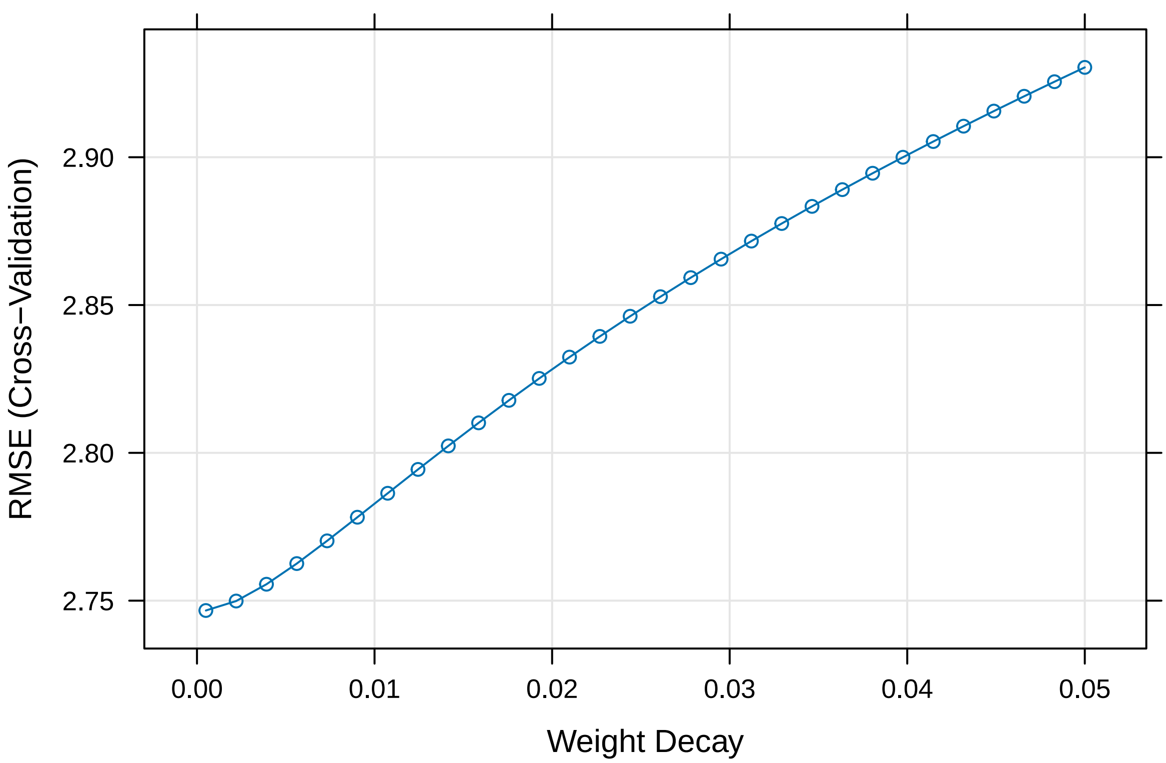
Data Splitting and Resampling Strategies

```
# Create training and test set data by using the oldest 80% of state-week observations as the training
flu_model_data1 <- flu_model_data1 %>%
  arrange(year, week)

cutoff <- floor(0.8 * nrow(flu_model_data1))

train_data <- flu_model_data1[1:cutoff, ]
test_data  <- flu_model_data1[(cutoff + 1):nrow(flu_model_data1), ]

# Subset the predictors into training and testing.
# First identify the response variables to remove them from the dataset.
target_vars <- c("target_1wk", "target_2wk", "target_3wk","target_4wk")

x_train <- train_data %>%
  dplyr::select(-all_of(target_vars))

x_test <- test_data %>%
  dplyr::select(-all_of(target_vars))

# Define the response variable as target_1wk since there are four target variables
# (i.e. target_1wk, target_2wk, target_3wk, target_4wk).
y_train <- train_data$target_1wk
y_test  <- test_data$target_1wk

# Create cross-validation folds
indx <- createFolds(y_train, k=5, returnTrain = TRUE)
ctrl <- trainControl(method = "cv", index = indx)
#head(train_data)
```

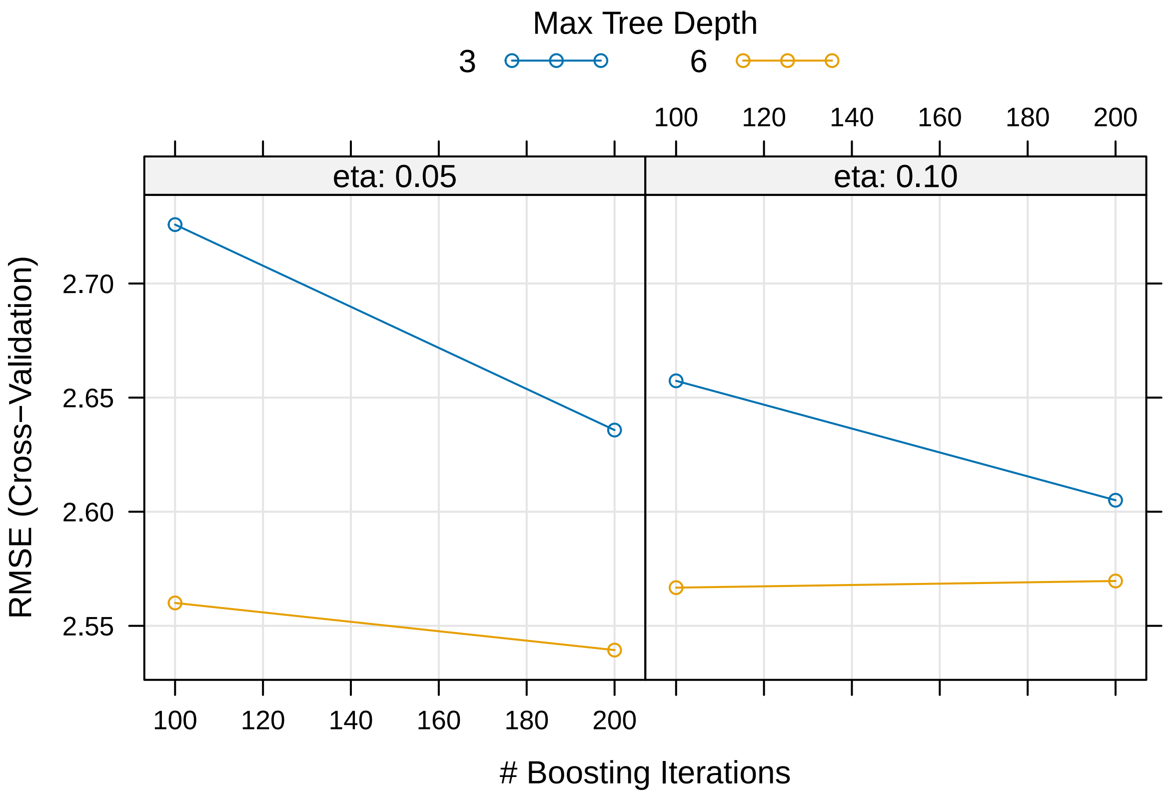
To create the training and test sets, instead of a stratified random sample or a simple random sample, a time-ordered sample was taken so that future observations did not leak into the training set given the time-series nature of influenza data. Next, the predictor set was created by removing all four response variables. The target variable, target_1wk was identified. Lastly, cross-validation folds were created where k=5 to reduce computation time.

Model Training and Tuning

```
# Fit the Model
# Ridge Regression
dummies <- dummyVars(~ ., data = x_train)

x_train_num <- predict(dummies, newdata = x_train)
x_test_num  <- predict(dummies, newdata = x_test)

set.seed(1)
ridgeGrid <- expand.grid(lambda = seq(0.0005, .05, length = 30))

set.seed(1)
ridgeTune <- train(x = x_train_num,
                   y = y_train,
                   method = "ridge",
                   tuneGrid = ridgeGrid,
                   trControl = ctrl
                   )

ridgeTune$bestTune
plot(ridgeTune)
```



```r
# Random Forest
# Specifying single value of mtry.
rfGrid <- expand.grid(
  mtry = floor(sqrt(ncol(x_train))))

set.seed(1)
rfTune <- train(x = x_train,
                y = y_train,
                method = "rf",
                tuneGrid = rfGrid,
                trControl = ctrl,
                importance = TRUE,
                ntree = 100
                )

# XGBoost
xgbGrid <- expand.grid(
  nrounds = c(100, 200),
  max_depth = c(3, 6),
  eta = c(0.05, 0.1),
  gamma = 0,
  colsample_bytree = 0.8,
  min_child_weight = 1,
  subsample = 0.8
)

set.seed(1)

xgbTune <- train(
  x = x_train_num,
  y = y_train,
  method = "xgbTree",
  tuneGrid = xgbGrid,
  trControl = ctrl
)

xgbTune$bestTune
plot(xgbTune)
```



For linear model evaluation, Ridge Regression was used due to its ability to handle highly correlated predictors. Random Forest was also chosen for its ability to reduce overfitting and for its high predictive accuracy. Lastly, XGBoost was selected for its ability to prevent overfitting and capture nonlinear relationships in the data.

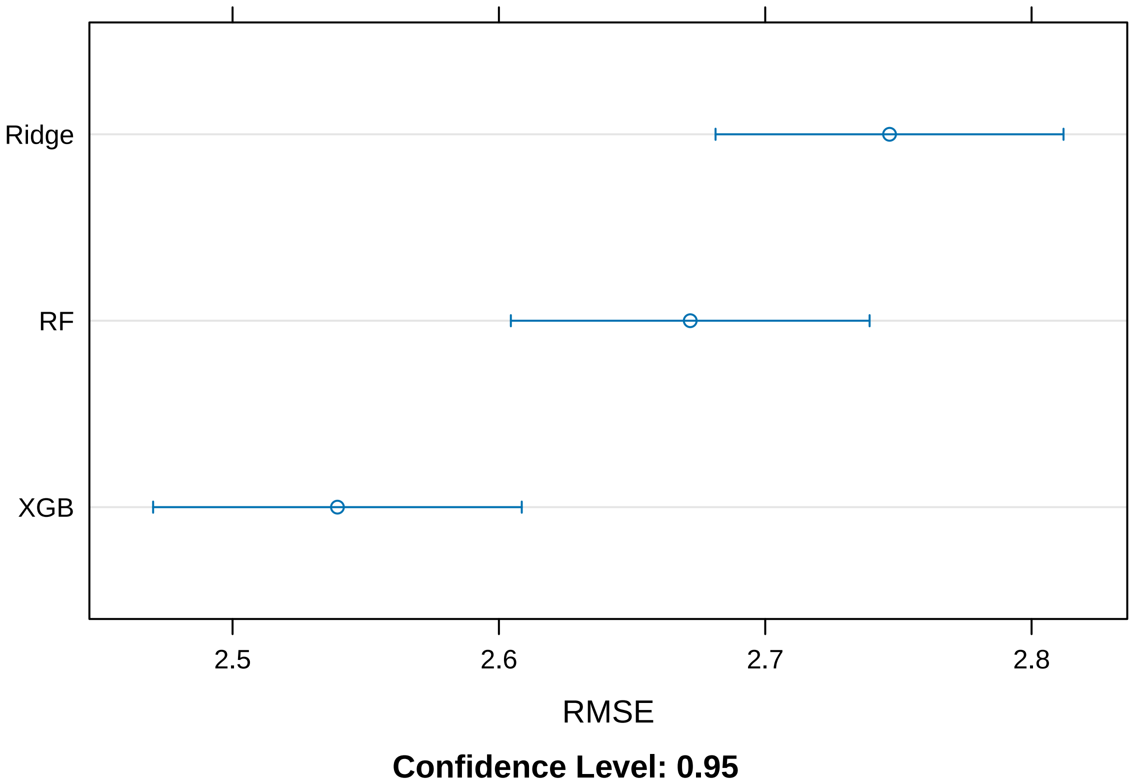
Validation, Tuning, and Testing

```r
# Model Evaluation
train_metrics <- resamples(list(Ridge = ridgeTune, RF = rfTune, XGB = xgbTune))
summary(train_metrics)
```

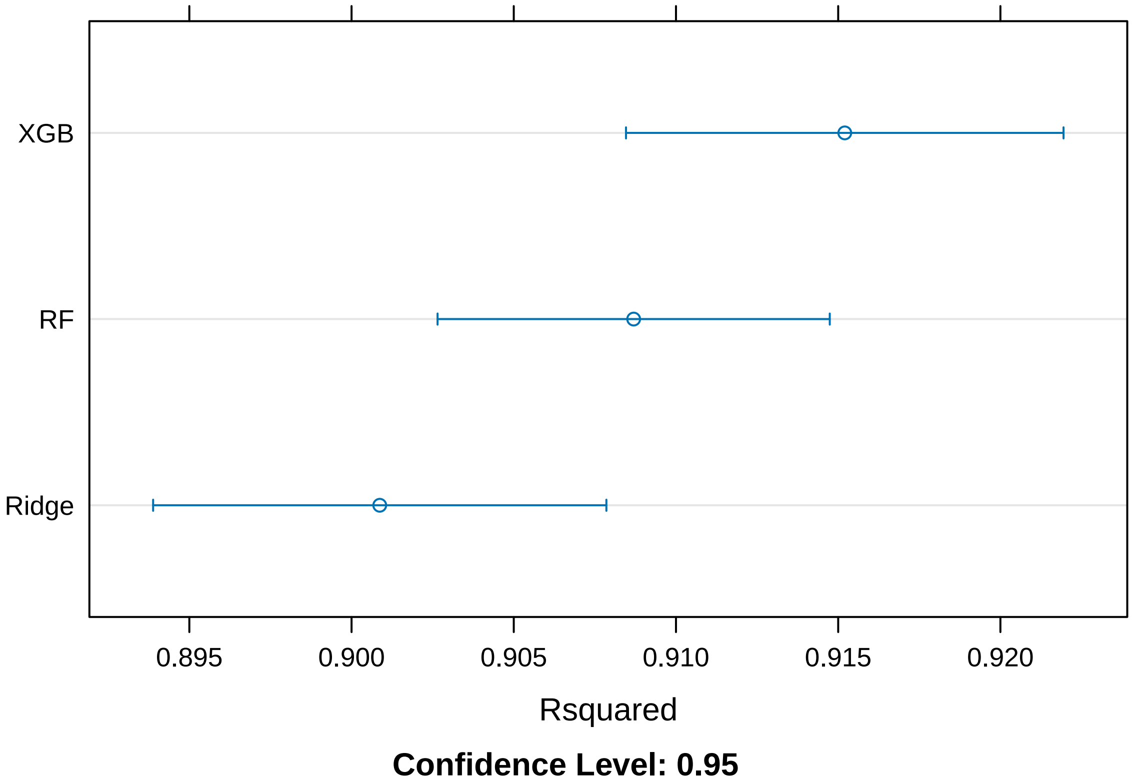
```
##
## Call:
## summary.resamples(object = train_metrics)
##
## Models: Ridge, RF, XGB
## Number of resamples: 5
##
## MAE
##           Min.  1st Qu.   Median     Mean  3rd Qu.     Max. NA's
## Ridge 1.549355 1.564260 1.567655 1.569719 1.578938 1.588385    0
## RF    1.398936 1.406084 1.407186 1.410657 1.412856 1.428221    0
## XGB   1.311331 1.322488 1.323841 1.328369 1.336987 1.347198    0
##
## RMSE
##           Min.  1st Qu.   Median     Mean  3rd Qu.     Max. NA's
## Ridge 2.691457 2.720946 2.724653 2.746673 2.770083 2.826226    0
```

```
## RF    2.607730 2.638601 2.659827 2.671839 2.711870 2.741167    0
## XGB   2.480365 2.504458 2.520407 2.539377 2.573551 2.618104    0
##
## Rsquared
##             Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## Ridge 0.8913175 0.9004686 0.9032976 0.9008693 0.9039196 0.9053431    0
## RF    0.9005928 0.9078429 0.9108820 0.9086959 0.9114847 0.9126772    0
## XGB   0.9066085 0.9140634 0.9169166 0.9152031 0.9172510 0.9211760    0
```

```r
dotplot(train_metrics, metric = "RMSE")
```




```r
dotplot(train_metrics, metric = "Rsquared")
```



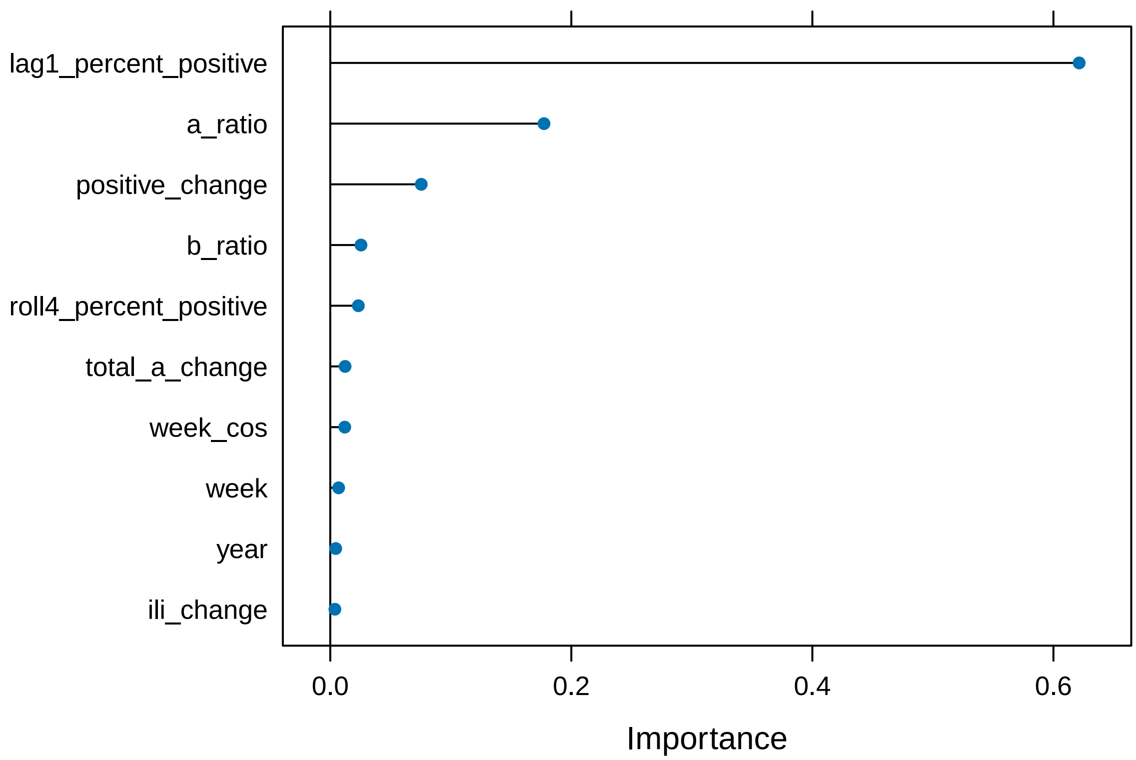

```r
# Determine variable importance
importance1 <- varImp(ridgeTune)
importance2 <- varImp(rfTune)
importance3 <- varImp(xgbTune)

ridge_imp <- data.frame(
  Variable = rownames(importance1$importance),
  Ridge = importance1$importance$Overall)

rf_imp <- data.frame(
  Variable = rownames(importance2$importance),
  RF = importance2$importance$Overall
)

xgb_imp <- data.frame(
  Variable = rownames(importance3$importance),
  XGB = importance3$importance$Overall
)

importance_table <- ridge_imp %>%
  full_join(rf_imp, by = "Variable") %>%
  full_join(xgb_imp, by = "Variable")

importance_table %>%
  arrange(desc(XGB)) %>%
  head(10) %>%
  knitr::kable(
    digits = 3,
```

```r
  caption = "Top 10 Variables by XGBoost Importance"
)
```

Table 26: Top 10 Variables by XGBoost Importance

Variable	Ridge	RF	XGB
lag1_percent_positive	100.000	72.760	100.000
a_ratio	89.241	55.436	28.541
positive_change	42.544	100.000	12.154
b_ratio	42.831	55.404	4.106
roll4_percent_positive	83.192	44.511	3.752
total_a_change	1.631	37.167	1.980
week_cos	21.068	55.470	1.934
week	7.124	60.359	1.119
year	1.810	44.307	0.728
ili_change	16.720	26.777	0.623

```r
xgbImp <- varImp(xgbTune, scale = FALSE)
plot(xgbImp, top = 10)
```




```r
# Save the test prediction results in a data frame
testResults <- data.frame(
  obs   = y_test,
  ridgePred = as.numeric(predict(ridgeTune,
                                  x_test_num)),
  rfPred = as.numeric(predict(rfTune,
                              x_test)),
```

```r
  xgbPred = as.numeric(predict(xgbTune,
                               x_test_num))
)

# Performance metrics on test set
test_metrics <- rbind(
  Ridge = postResample(pred = testResults$ridgePred, obs = testResults$obs),
  RF    = postResample(pred = testResults$rfPred, obs = testResults$obs),
  XGB   = postResample(pred = testResults$xgbPred, obs = testResults$obs)
)

test_metrics
```

```
##           RMSE  Rsquared      MAE
## Ridge 2.618860 0.9071010 1.450931
## RF    2.772335 0.9028696 1.535733
## XGB   2.384147 0.9247554 1.290796
```

```r
# Check for statistical significance in model performance differences
diff(train_metrics, metric = "RMSE") |> summary()
```

```
##
## Call:
## summary.diff.resamples(object = diff(train_metrics, metric = "RMSE"))
##
## p-value adjustment: bonferroni
## Upper diagonal: estimates of the difference
## Lower diagonal: p-value for H0: difference = 0
##
## RMSE
##       Ridge     RF        XGB
## Ridge           0.07483   0.20730
## RF    0.0005259           0.13246
## XGB   6.516e-05 0.0004032
```

A check for statistically significant differences in the RMSE values of each of the models reveals that there is a significant difference in RMSE between XGB and Ridge (p-value = 0.00004) and between XGB and RF (p-value = 0.020). The difference in the prediction error between Ridge and RF (p-value = 0.232) was not statistically significant. XGB achieved the lowest RMSE (2.384) and highest R-squared with a value of 0.925, and, as a result, was selected as the preferred model for prediction.